

Security Lab Solution Document

Observability and Response Architecture Pt.1



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Contents

Observability and Response Architecture	2
Splunk Enterprise SIEM Platform	4
Log Aggregation.....	6
Data Normalization and Field Extraction	6
Elastic Stack (ELK) Observability Platform.....	12
Component Architecture:	13
Core ELK Stack Configuration	13
Distributed Elastic Agents.....	14
Kibana Dashboards	16
Wazuh Endpoint Detection and Response (EDR)	22
Component Architecture	23
Security Monitoring Modules:	26
Active Response	28
Integrations	30
Alerting and Notifications	35
Wazuh Dashboard Analytics:	35

Observability and Response Architecture

Architecture Overview

The lab implements a comprehensive observability and security monitoring stack that provides real-time visibility across infrastructure, applications, and security events. This multi-layered approach demonstrates enterprise-grade security operations center (SOC) capabilities and incident response readiness.

Core Technology Stack:

- SIEM Platforms: Splunk Enterprise (primary), Elastic Stack (secondary)
- Endpoint Detection: Wazuh EDR agents deployed on 25+ hosts across Windows, Linux, BSD, and macOS
- Network Monitoring: Suricata/Snort IDS, firewall logging, Network Sensor (Zeek/ntopng), NetAlertx
- Infrastructure Monitoring: Prometheus + Grafana for metrics visualization, CheckMK for system monitoring
- Service Availability: Uptime Kuma for service availability tracking, Pulse for Proxmox-specific monitoring
- Alerting Hub: Discord notifications through webhooks and email alerts through local SMTP relay.

Deployment Rationale:

Security visibility is foundational to detecting and responding to threats before they cause damage. This architecture demonstrates:

- **Correlation across multiple data sources:** Network, host, and application telemetry
- **Real-time detection:** Security events and anomalies identified within seconds
- **Hands-on experience:** Industry-standard SIEM and monitoring tools (Splunk, Elastic, Prometheus, Wazuh)
- **Enterprise SOC patterns:** Integration workflows used in production environments
- **Automated response:** Incident response workflows triggered by alert conditions

Architecture Principles Alignment:

- **Defense in Depth:** Real-time SIEM correlation across multiple data sources (network, host, application); multiple detection mechanisms reduce blind spots (SIEM, EDR); correlation across network, application, and endpoint layers; redundant monitoring platforms (Splunk + Elastic) ensure continuity
- **Secure by Design:** Integration patterns demonstrate enterprise SOC workflows; documented playbooks and alert workflows for incident response
- **Zero Trust:** Continuous verification through comprehensive logging (100% security event coverage); Identity-aware monitoring tracks user behavior across systems

Strategic Value:

Comprehensive Visibility:

- 360-degree view from network perimeter to application layer
- Unified dashboards provide single pane of glass for security posture

Defense in Depth:

- Multiple detection mechanisms reduce single points of failure
- Overlapping coverage ensures threats are caught by at least one control

Skills Development:

- Experience with commercial (Splunk) and open-source (Elastic) platforms
- Hands-on with tools directly transferable to SOC Analyst roles

Incident Response:

- Documented playbooks and alert workflows demonstrate operational maturity
- Integration with TheHive enables case management and forensic tracking

Compliance Readiness:

- Audit logs retained for 90 days with searchable correlation
- Immutable SIEM indexes support forensic investigations
- Comprehensive logging satisfies PCI-DSS 10.2, NIST SP 800-53 AU-2

Why Multiple SIEM Platforms?

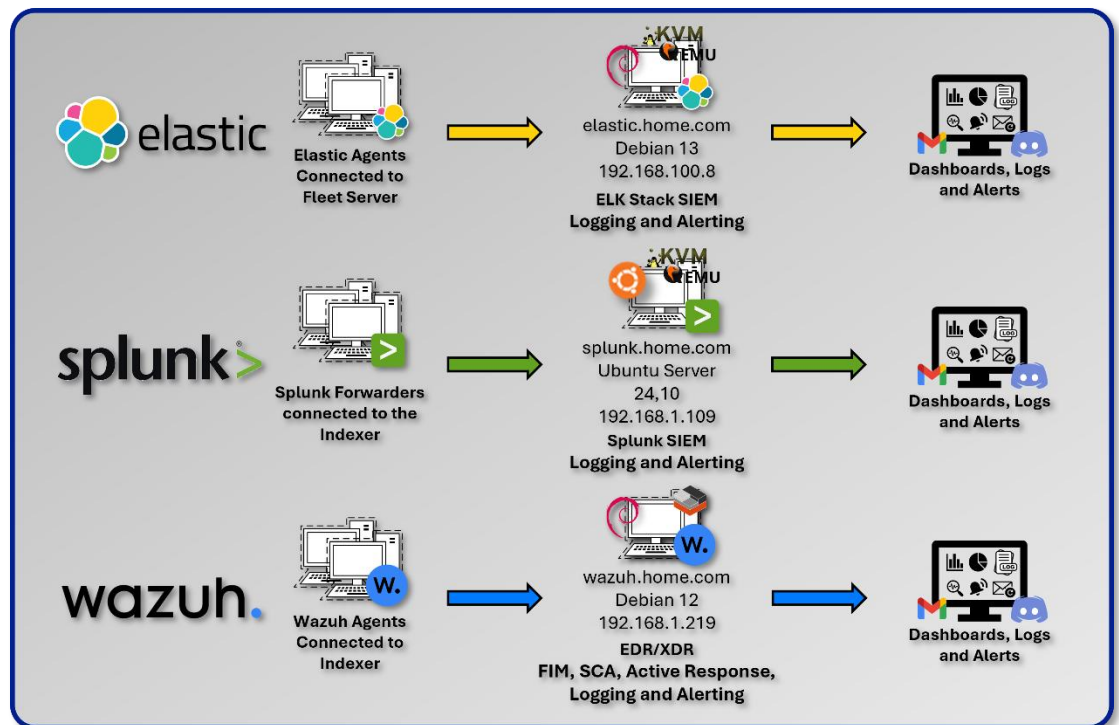
Splunk (Primary):

- Mature SPL query language for security event correlation
- Industry-standard platform used in enterprise SOCs
- Robust alerting and dashboard capabilities

Elastic (Secondary):

- ECS (Elastic Common Schema) standardization for vendor-neutral data modeling
- Focus on application telemetry and infrastructure metrics
- Open-source alternative demonstrates platform versatility

Rationale: Dual deployment provides redundancy and learning opportunities with both commercial and open-source platforms commonly used in enterprise environments. This approach demonstrates adaptability and reduces vendor lock-in risk.



Splunk Enterprise SIEM Platform Deployment Overview

Splunk Enterprise functions as the centralized Security Information and Event Management (SIEM) platform for the lab environment, providing unified log aggregation, correlation, and real-time detection across all security and infrastructure layers. The deployment uses a single-instance architecture optimized for lab-scale operations, with a clear expansion path toward distributed indexing and search

head clustering as data volume grows. Splunk ingests telemetry from firewalls, IDS/IPS sensors, EDR agents, authentication systems, and application logs, normalizing events through the Common Information Model (CIM) to enable consistent, cross-source analytics.

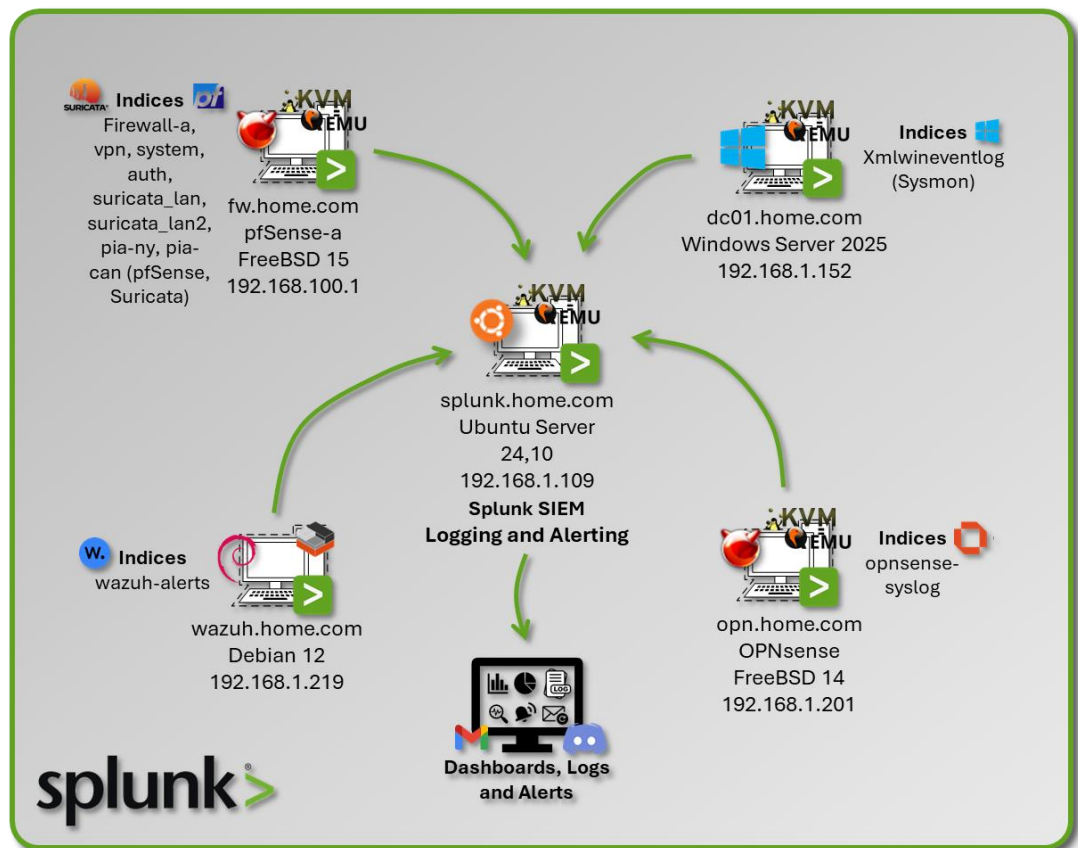
Security Impact

Splunk delivers enterprise-grade detection and visibility by consolidating more than 30 data sources into a single analytical plane. Real-time alerting identifies brute-force attempts, port scans, anomalous authentication behavior, and early indicators of lateral movement. Correlation searches detect multi-stage attack patterns such as privilege escalation, internal reconnaissance, and data exfiltration. Redundant detection paths across pfSense, Suricata, and Wazuh sourcetypes ensure adversary activity is captured even if one telemetry source is degraded. Universal Forwarders automate log collection at scale, while scheduled alerts trigger automated incident-response workflows. Integration with Discord provides immediate notification channels, and SOC-style dashboards offer continuous operational visibility.

Deployment Rationale

A SIEM is the analytical backbone of any modern security program, enabling defenders to transform raw telemetry into actionable intelligence. Without centralized log aggregation and correlation, organizations lack the ability to detect distributed attacks, validate security control effectiveness, or perform meaningful threat hunting. This Splunk deployment demonstrates core SIEM competencies found in enterprise environments: CIM-based normalization, multi-source correlation, automated alerting, and integration with downstream SOAR workflows. The architecture reflects real-world SOC operations where SIEM platforms serve as the authoritative source of truth for incident detection, investigation, and compliance reporting.

Architecture Principles Alignment



Defense in Depth

- Aggregates telemetry from every security layer (network, endpoint, identity, application) to create overlapping detection coverage
- Correlation searches combine multiple signals to detect attacks that bypass single controls
- Redundant sourcetypes (pfSense, Suricata, Wazuh) ensure detection continuity even if one sensor fails

Secure by Design

- CIM standardization enforces consistent field structures, reducing parsing errors and blind spots
- Universal Forwarders provide authenticated, encrypted log transport
- Automated alerting and workflow triggers reduce human error and accelerate response

Zero Trust

- No event or data source is implicitly trusted; all logs are validated, normalized, and correlated
- Identity-centric analytics detect anomalous authentication and privilege misuse
- Continuous monitoring ensures every action, user, and system interaction is observable and attributable

Deployment Architecture:

Component	Technology	Location	Purpose
Splunk Indexer	Splunk Enterprise 9.2.x	Ubuntu VM	Log storage and search
Splunk Forwarder	Universal Forwarder 9.2	All monitored hosts	Log collection and forwarding
Data Inputs	Syslog	file monitoring	Various Ingestion methods
Indexes	Time-series buckets	/opt/splunk/var/lib	Segmented storage

Universal Forwarder Deployment:

Host Type	Forwarder Count	Logs Forwarded	Volume/Day
pfSense Firewalls	2	Firewall, VPN, System, Auth	500 MB
OPNsense Firewall	1	Firewall logs	100 MB
Suricata IDS	4 (Internal LANs)	IDS/IPS alerts flow data	2.5 GB
Snort IDS	2 (VPN tunnels)	IDS alerts	200 MB
Wazuh Manager	1	Security alerts audit logs	1 GB
Windows Hosts	1 (Sysmon), 4 (Wazuh)	Sysmon XML events, Wazuh	800 MB (Sysmon)
Linux Hosts	15 (Wazuh)	Wazuh	
MacOS Hosts	1 (Wazuh)	Wazuh	

Index Design:

Index Name	Data Sources	Retention	Size/Day	Purpose
firewall-a	pfSense filterlog	90 days	400 MB	Network traffic logs
vpn	pfSense VPN logs	90 days	50 MB	VPN connection logs
system	pfSense system logs	90 days	50 MB	System events
auth	pfSense auth logs	90 days	20 MB	Authentication events
opnsense-syslog	OPNsense firewall	90 days	100 MB	Firewall logs
suricata-prod_lan	Suricata Prod_LAN interface	90 days	1 GB	IDS alerts (internal)
suricata-lab_lan1	Suricata Lab_LAN1 interface	90 days	.5 GB	IDS alerts (external)
suricata-lab_lan2	Suricata Lab_LAN2 interface	90 days	.5 GB	IDS alerts (external)
suricata-ext_lan	Suricata Ext_LAN interface	90 days	.5 GB	IDS alerts (external)
pia-ny	Snort on PIA NY tunnel	90 days	100 MB	VPN tunnel IDS
pia-can-mont	Snort on PIA Montreal	90 days	100 MB	VPN tunnel IDS
wazuh-alerts	Wazuh security events	90 days	1 GB	Endpoint security
xmlwineventlog	Windows Sysmon	90 days	800 MB	Windows telemetry

Index Optimization:

- Hot/Warm/Cold Architecture: 7 days hot, 30 days warm, 53 days cold
- Compression: ~70% compression ratio on warm/cold buckets
- Frozen: Archive to NAS after 90 days (compressed tar.gz)

Log Aggregation

Data Inputs:

Source	Input Type	Method	Notes
Wazuh	JSON over TCP	Splunk Universal Forwarder	Includes security alerts and audit logs
pfSense	Log files	Splunk Universal Forwarder	Firewall logs, VPN, authorization, and System.
Suricata	JSON over File	Splunk Universal Forwarder	IDS/IPS alerts, flow data
OPNsense	Syslog (UDP)	Splunk Universal Forwarder	Firewall logs
Sysmon	XML via EventLog	Splunk Universal Forwarder	Windows endpoint telemetry, process tracking

Data Normalization and Field Extraction

Common Field Mapping:

Common Information Model (CIM) Compliance:

Splunk CIM provides standardized field names across data sources for correlation.

CIM Data Model	Normalized Fields	Source Types
Network Traffic	src_ip, dest_ip, src_port, dest_port	pfsense:filterlog, suricata
Authentication	user,src, dest, action	pfsense:auth, wazuh-alerts
Intrusion Detection	signature, severity, category	suricata, snort
Endpoint	process, parent_process, user	xmlwineventlog:sysmon

Field Extraction Example (transforms.conf):

```
[pfsense-filterlog-extract]
REGEX =
^(\S+)\s+\S+\s+filterlog\[\\d+\\]:\\s+\\d+,,, (\\d+), (\\w+), (\\d+), (\\S+), (\\d+), (\\S+), (\\d+), (\\w+), (\\d+), (\\d+), (\\d+), (\\S+), (\\d+), (\\d+), (\\d+), (\\w+)
FORMAT = interface::$1 rule_number::$2 action::$3 protocol::$4 src_ip::$5 src_port::$6 dest_ip::$7 dest_port::$8
```

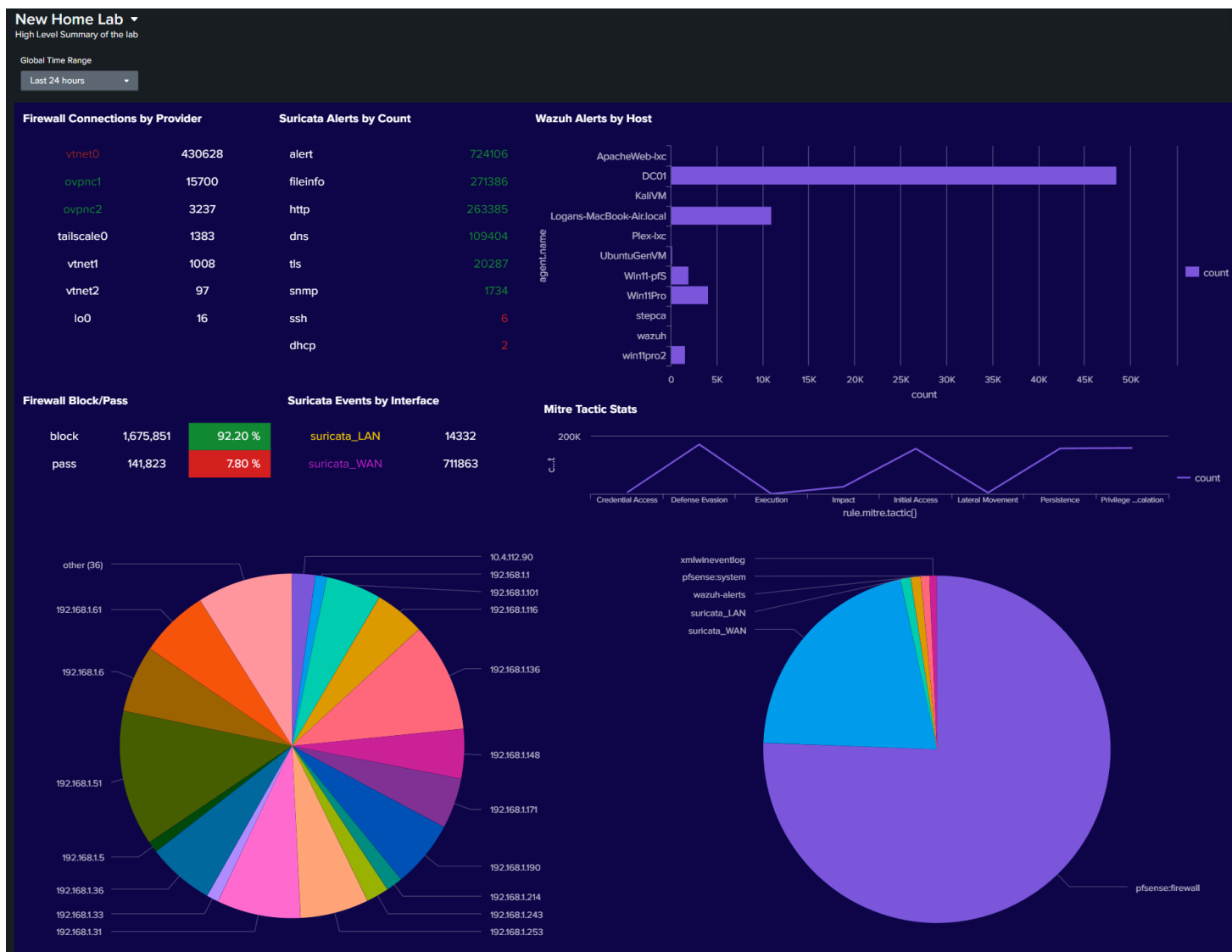
Searches, Reports, and Dashboards

Security Operations Center (SOC) Dashboards:

SIEM Overview Dashboard:

Panels:

- **Firewall Connections by Provider:** A chart/table that lists network interfaces (providers) like vtnet0, vtnet1, tailscale0, etc., alongside the count of connections associated with each.
- **Suricata Alerts by Count:** Lists specific types of Suricata security alerts (alert, fileinfo, http, dns, tls, snmp, ssh, dhcp) and their frequency (count). alert has the highest count at 430,628.
- **Wazuh Alerts by Host:** A horizontal bar chart showing the count of alerts generated by specific hosts or agents (ApacheWeb-tec, DC01, KVM, etc.). ApacheWeb-tec has the highest alert count at over 40,000.
- **Firewall Block/Pass:** A status indicator showing the total count and percentage of blocked (1,675,851 / 92.20%) versus passed traffic (141,823 / 7.80%).
- **Suricata Events by Interface:** A colorful pie chart visualizing which internal IP addresses or network segments (192.168.x.x ranges) are generating the most Suricata security events.
- **Mitre Tactic Stats:** A line/area chart tracking the count of events mapped to different MITRE ATT&CK Tactics over time, such as Defense Evasion, Impact, Lateral Movement, etc.
- **Services Pie Chart:** A large pie chart showing the distribution of traffic categorized by the application or service, likely derived from flow logs or firewall data.



Wazuh alerts for my Windows Server 2022 Domain Controller.

Index=wazuh-alerts, Agent.name=DC01

New Search Save As Create Table View Close

Index=wazuh-alerts "agent.name"=DC01
| stats count by rule.description, data.win.eventdata.subjectUserName
| sort - count

174,544 events (1/25 12:00:00.000 AM to 8/23/25 8:08:58.000 PM) No Event Sampling

Events Patterns **Statistics (7)** Visualization

Log Message	Agent Name
Special privileges assigned to new logon.	superpaul
Non network or service local logon.	DC01\$
Special privileges assigned to new logon.	DWM-2
User: lab\superpaul logged using Remote Desktop Connection (RDP) from ip:192.168.1.31.	DC01\$
Special privileges assigned to new logon.	DWM-2
User: lab\superpaul logged using Remote Desktop Connection (RDP) from ip:192.168.1.31.	DC01\$

Intrusion Prevention System alerts for Suricata

Index=ids,sourcetype=suricata

New Search

index=ids sourcetype="suricata" | stats count by signature, src_ip, dest_ip, severity, _time | sort - count

85,973 events (1/1/25 12:00:00.000 AM to 8/23/25 7:43:38.000 PM) No Event Sampling

EventsPatternsStatistics (10,000)Visualization

Show: 50 Per PageFormatPreview: On

Prev

signature	src_ip	dest_ip	severity	_time
SURICATA STREAM excessive retransmissions	208.123.73.207	192.168.1.253	low	2025-08-05 13:59:57.092
SURICATA ICMPv4 invalid checksum	192.168.1.253	192.168.1.249	low	2025-08-05 13:36:42.935
SURICATA ICMPv4 invalid checksum	192.168.1.253	192.168.1.249	low	2025-08-05 13:36:51.514
SURICATA ICMPv4 invalid checksum	192.168.1.253	192.168.1.249	low	2025-08-05 14:04:01.672
SURICATA ICMPv4 invalid checksum	192.168.1.253	192.168.1.249	low	2025-08-05 14:04:03.719
SURICATA ICMPv4 invalid checksum	192.168.1.253	192.168.1.249	low	2025-08-05 14:04:05.765

SURICATA ICMPv4 invalid checksum192.168.1.253192.168.1.249low

SURICATA QUIC failed decrypt192.168.1.25323.43.85.13low

SURICATA QUIC failed decrypt192.168.1.25323.43.85.13low

SURICATA ICMPv4 invalid checksum	192.168.1.253	192.168.1.249	low	2025-08-05 15:16:06.334
SURICATA QUIC failed decrypt	192.168.1.253	23.43.85.13	low	2025-08-05 02:51:55.824
SURICATA QUIC failed decrypt	192.168.1.253	23.43.85.13	low	2025-08-05 02:51:55.826
SURICATA QUIC failed decrypt	192.168.1.253	23.43.85.13	low	2025-08-05 02:51:55.827
SURICATA QUIC failed decrypt	192.168.1.253	23.43.85.39	low	2025-08-05 02:46:10.144
SURICATA QUIC failed decrypt	192.168.1.253	23.43.85.39	low	2025-08-05 02:46:10.145
SURICATA QUIC failed decrypt	192.168.1.253	23.45.233.39	low	2025-08-05 02:51:55.906
SURICATA QUIC failed decrypt	192.168.1.253	23.45.233.39	low	2025-08-05 02:51:55.910
SURICATA QUIC failed decrypt	192.168.1.253	23.45.233.39	low	2025-08-05 02:51:55.911
SURICATA QUIC failed decrypt	192.168.1.253	23.49.251.211	low	2025-08-05 02:51:54.896
SURICATA QUIC failed decrypt	192.168.1.253	23.49.251.211	low	2025-08-05 02:51:54.898
SURICATA QUIC failed decrypt	192.168.1.253	23.49.251.211	low	2025-08-05 02:51:54.922
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 02:14:38.916
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 02:24:45.380
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 02:29:55.512
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 02:34:59.514
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 02:40:03.514
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 03:01:56.027
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 03:20:06.985
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 03:35:28.927
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 03:52:52.025
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 04:09:17.910
ET POLICY Spotify P2P Client	192.168.1.106	192.168.1.255	low	2025-08-05 04:24:53.000

Firewall filter log for pfSense

Index=firewall-a, sourcetype=pfsense:firewall

New Search

index=* OR index=* sourcetype=pfsense:firewall | table src_ip, dst_ip, src_port, action

954,431 events (8/22/25 8:00:00.000 PM to 8/23/25 8:24:36.000 PM) No Event Sampling Job

Events Patterns Statistics (954,431) Visualization

Show: 50 Per Page Format Preview: On

src_ip	dst_ip	src_port	action
192.168.100.51	192.168.1.247	34292	pass
192.168.100.51	192.168.1.250	40143	pass
192.168.100.51	192.168.1.247	57310	pass
192.168.100.51	192.168.1.250	46978	pass
192.168.100.51	192.168.1.250	55224	pass
192.168.100.51	192.168.1.250	58481	pass
192.168.100.51	192.168.1.247	39018	pass
192.168.100.51	192.168.1.250	49609	pass
192.168.100.51	192.168.1.247	52040	pass
192.168.100.51	192.168.1.250	32942	pass
192.168.100.15	173.68.122.128	41938	pass
192.168.100.15	192.168.1.250	44789	pass
192.168.100.15	173.68.122.128	44976	pass
192.168.100.15	192.168.1.250	69547	pass
192.168.100.15	173.68.122.128	44972	pass
192.168.100.15	192.168.1.250	47406	pass
192.168.100.15	173.68.122.128	52218	pass
192.168.100.15	173.68.122.128	50622	pass
192.168.100.15	192.168.1.250	50215	pass
192.168.100.15	173.68.122.128	33582	pass
192.168.100.15	192.168.1.250	58185	pass
192.168.100.15	173.68.122.128	54798	pass
192.168.100.15	192.168.1.250	58151	pass
192.168.100.15	173.68.122.128	34370	pass
192.168.100.15	173.68.122.128	34830	pass
192.168.100.15	192.168.1.250	36981	pass
192.168.100.15	173.68.122.128	58928	pass

Searches:

Brute Force Detection (Multi-Source):

```
(index=auth sourcetype=pfsense:auth Failed password) OR
(index=wazuh-alerts rule.description=*authentication failed*)
| eval src_ip=coalesce(src,source_ip)
| stats count by src_ip, dest
| where count > 10
| eval threat_level=High
```

Port Scan Detection:

```
index=firewall-a action=block
| stats dc(dest_port) as unique_ports, count by src_ip, dest_ip
| where unique_ports > 50 AND count > 100
| eval description=Possible port scan: .src_ip. scanning
.unique_ports. ports on .dest_ip
```

VPN Traffic Analysis:

```
index=firewall-a sourcetype=pfsense:filterlog
| eval interface=if(like(interface,%ovpnc%),VPN,Direct)
| stats sum(bytes) as total_bytes by interface
| eval total_gb=round(total_bytes/1024/1024/1024,2)
```

Alert Configuration:

Scheduled Alerts:

Alert Name	Search Schedule	Trigger Condition	Action
Failed SSH Brute Force	Every 5 minutes	>20 failed attempts	Discord/Gmail
Critical Wazuh Alert	Real-time	rule.level >= 12	Discord/Gmail
Suricata High Severity	Every 15 minutes	alert.severity = 1	Discord/Gmail
VPN Tunnel Down	Every 1 minute	No VPN traffic in 2 min	Discord/Gmail
Anomalous Outbound Traffic	Every 30 minutes	Bytes > 10GB in 30min	Discord/Gmail

Alert Tuning and False Positive Management:

- Baseline Period: 30 days to establish normal behavior
- Dynamic Thresholds: Adjust based on time of day, day of week
- Whitelist: Known benign IPs/hosts excluded from alerts
- Alert Suppression: Maximum 1 alert per source per hour
- Escalation: Critical alerts escalate after 15 minutes if unacknowledged

Security Use Cases:

Use Case 1: Lateral Movement Detection

Objective: Detect unauthorized account access across multiple systems

SPL Query:

```
index=wazuh-alerts OR index=auth
| eval user=coalesce(user.name,user)
| stats dc(agent.name) as unique_hosts by user
| where unique_hosts > 5
| eval threat=Potential lateral movement: user accessed .unique_hosts. Hosts
```

Use Case 2: Data Exfiltration Detection

Objective: Identify large outbound data transfers

SPL Query:

```
index=firewall-a action=pass
| eval outbound=if(like(dest_ip,192.168.%),0,1)
| where outbound=1
| stats sum(bytes) as total_bytes by src_ip, dest_ip
| where total_bytes > 1073741824
| eval total_gb=round(total_bytes/1024/1024/1024,2)
| eval alert=Potential data exfiltration: .total_gb.GB transferred
```

Use Case 3: Malware Execution Detection

Objective: Detect execution of known malicious files

SPL Query:

```
index=xmlwineventlog EventCode=1
| eval md5=lower(Hashes)
| lookup malware_hashes.csv md5 OUTPUT malware_name
| where isnotnull(malware_name)
| table _time, Computer, User, Image, malware_name, CommandLine
```

Elastic Stack (ELK) Observability Platform

Deployment Overview



The Elastic Stack provides end-to-end observability across applications, infrastructure, and security events using the Elastic Common Schema (ECS) for standardized data modeling.

This deployment emphasizes application performance monitoring (APM), infrastructure telemetry, and container visibility, complementing Splunk's security-centric analytics. Fleet

Server centrally manages more than 25 distributed Elastic Agents, enforcing consistent policies across hybrid environments including VMs, Docker containers, and Kubernetes workloads. Kibana dashboards deliver real-time insights into system performance, application behavior, and operational health.

Security Impact

Elastic enhances the security posture by delivering deep visibility into application and infrastructure behavior. ECS-compliant data modeling enables vendor-neutral threat detection and correlation across diverse telemetry sources. Infrastructure metrics expose resource exhaustion attacks, while container security monitoring identifies anomalous behavior within Docker and Kubernetes environments.

High-cardinality metrics support granular detection of performance anomalies, and integration with Prometheus/Grafana demonstrates multi-tool observability proficiency. Kibana's analytics capabilities empower developers and operators with self-service dashboards for rapid troubleshooting and threat validation.

Deployment Rationale

While Splunk serves as the primary SIEM for security event correlation, Elastic Stack provides complementary strengths in application-level observability, infrastructure telemetry, and open-source analytics. ECS-based normalization supports vendor-agnostic correlation and multi-SIEM workflows. This deployment demonstrates proficiency with modern observability ecosystems, including APM instrumentation, container monitoring, and hybrid infrastructure analytics. Elastic's open-source model showcases the ability to operate and integrate multiple analytical platforms within a unified security and observability strategy.

Architecture Principles Alignment

Defense in Depth

- Adds application-level and infrastructure-level visibility to complement network and endpoint telemetry
- Container and Kubernetes monitoring provide additional layers of behavioral detection
- ECS normalization enables correlation across heterogeneous data sources

Secure by Design

- ECS enforces consistent field structures for reliable analytics and reduced parsing errors
- Fleet Server centralizes agent policy enforcement and reduces configuration drift

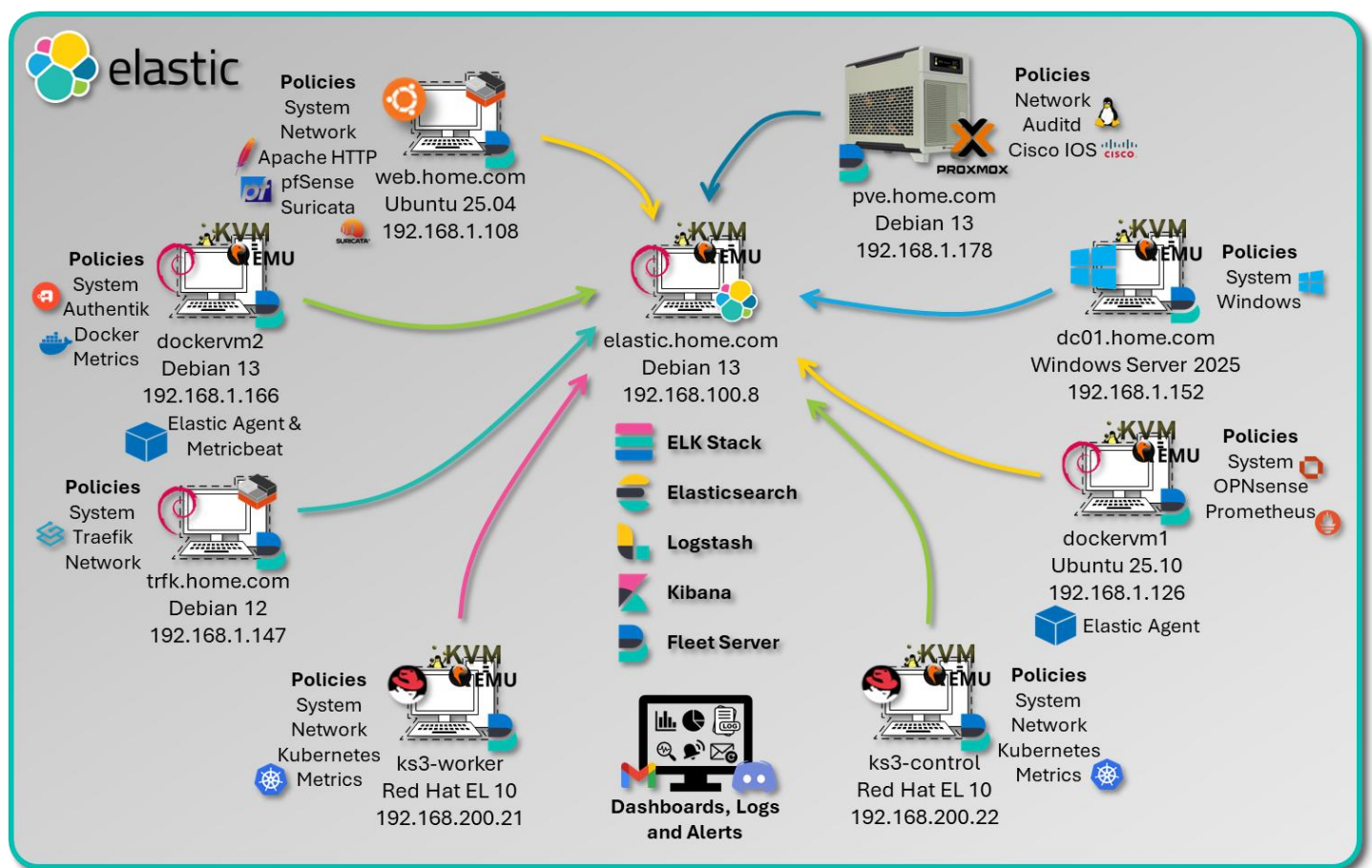
- Scalable architecture supports high-cardinality metrics for precise anomaly detection

Zero Trust

- No telemetry source is implicitly trusted; all events are normalized and validated through ECS
- Continuous monitoring of applications, containers, and infrastructure ensures full observability
- Segmented data pipelines prevent cross-contamination between workloads and environments

Component Architecture:

- **Elasticsearch:** A distributed search and analytics engine that stores and indexes structured and unstructured data.
- **Logstash:** A data processing pipeline that ingests, transforms, and forwards logs to Elasticsearch (used selectively in this lab).
- **Kibana:** A visualization and management interface for exploring data, building dashboards, and managing Fleet and integrations.
- **Fleet Server:** Manages Elastic Agent enrollment, policy distribution, and secure communication with Elasticsearch.



Core ELK Stack Configuration

The central Elastic Stack is deployed on a **KVM virtual machine running Debian 13**, hosting the following components:

- Elasticsearch, Logstash, and Kibana (ELK)
- Fleet Server and a local Elastic Agent
- Version: 9.2.5

This VM serves as the central ingest, processing, and visualization hub for all telemetry across the lab environment.

Distributed Elastic Agents

Elastic Agents are deployed across multiple hosts, each assigned a tailored **Fleet policy** with relevant integrations to ensure ECS-compliant data collection and enrichment:

Hostname	OS / Platform	Agent Policy Integrations
Traefik LXC Host	LXC container	system, Traefik integration for access logs and metrics, Network Packet Logs
DC01	Windows Server 2022 VM	system, Windows, and Sysmon integrations for endpoint and process telemetry
DockerVM1	Ubuntu VM	system, Prometheus, and OPNSense (pfSense) integrations
DockerVM2	Docker container running on Debian VM	system, Docker, authentik integration for event and user activity logs
Proxmox PVE Host	Proxmox VE (Debian)	system, Auditd integration for kernel-level audit logging, Cisco iOS Syslog
Ubuntu-Apache LXC	LXC container	system, Apache HTTP Server, Suricata, and pfSense integrations for web and IDS logs
K3s-control	Kubernetes container	System, Kubernetes metrics, network packet capture
K3s-worker	Kubernetes container	System, Kubernetes metrics, network packet capture

Fleet

Centralized management for Elastic Agents.

[Agents](#) [Agent policies](#) [Enrollment tokens](#) [Uninstall tokens](#) [Data streams](#) [Settings](#)

Ingest Overview Metrics

Agent Info Metrics

⌚ Agent Info

🔍 Filter your data using KQL syntax

Status 5 ▼ Tags 1 ▼

Showing 9 agents ⓘ [Clear filters](#)

● Healthy 9 ● Unhealthy 0 ● Orphaned 0 ● Updating 0 ● Offline 0

☐	Status	Host ↕	Agent policy ↕	CPU ⓘ	Memory ⓘ	Last activity ↕	Version ↕
☐	Healthy	5f21af39bf66	DockerVM2 Agent Policy rev. 11	0.47 %	292 MB	20 seconds ago	9.2.2 ⓘ
☐	Healthy	k3-control	Agent policy 7 rev. 8	1.68 %	668 MB	14 seconds ago	9.2.2 ⓘ
☐	Healthy	k3-worker	Agent policy 7 rev. 8	0.99 %	510 MB	11 seconds ago	9.2.2 ⓘ
☐	Healthy	traefik	Traefik Agent Policy rev. 14	0.84 %	634 MB	10 seconds ago	9.2.3
☐	Healthy	DC01.home.com	DC01 Agent Policy rev. 15	N/A ⓘ	N/A ⓘ	11 seconds ago	9.2.2
☐	Healthy	UbuntuVM1	DockerVM1 Agent Policy rev. 26	1.14 %	425 MB	5 seconds ago	9.2.2
☐	Healthy	pve	pve Agent Policy rev. 19	7.79 %	1 GB	23 seconds ago	9.2.2
☐	Healthy	apache-ubuntu.home.com	Apache Agent Policy rev. 26	0.78 %	677 MB	8 seconds ago	9.2.2
☐	Healthy	elastic	Fleet Server policy 1 rev. 10	0.72 %	250 MB	5 seconds ago	9.2.3

Each agent is enrolled via Fleet and centrally managed through Kibana, enabling consistent policy updates, secure API key rotation, and streamlined observability across the lab.

Fleet Policy Configuration:

Example: Windows Endpoint Policy

Integrations:

- System: CPU, memory, disk, network metrics
- Windows: Event logs (Security, Application, System)
- Sysmon: Process creation, network connections, registry changes

Data Stream Configuration:

logs-windows.sysmon-default:

- EventID 1: Process creation
- EventID 3: Network connection
- EventID 7: Image loaded
- EventID 10: Process access
- EventID 11: File creation
- EventID 13: Registry value set

metrics-system.cpu-default:

- Metricset: cpu
- Period: 10s
- CPU percentages: user, system, idle, iowait

Elastic Common Schema (ECS) Mapping

ECS Field Standardization:

The lab leverages ECS to normalize fields across heterogeneous data sources, enabling unified dashboards and cross-source correlation.

Integration	ECS Dataset Prefix	Key ECS Fields Used
System	system.*	host.name, process.name, user.name, event.type
Windows/Sysmon	windows.*	event.code, process.parent.name, registry.path
Docker	docker.*	container.id, container.image.name, host.hostname
Prometheus	prometheus.*	metricset.name, service.name, labels.*
Suricata	suricata.eve	network.protocol, source.ip, event.category
Apache	apache.access	http.request.method, url.path, user_agent.name
Authentik	authentik.* (custom)	user.name, event.action, event.dataset
Auditd	auditd.*	process.executable, event.action, user.id
pfSense/OPNSense	firewall.*	source.ip, destination.port, event.outcome

ECS Benefits:

- Vendor Neutrality: Same queries work across different data sources
- Future-Proof: Adding new integrations automatically compatible

- **Community Standard:** Widely adopted in security and observability communities

Custom ingest pipelines ensure ECS compliance for integrations like Authentik and pfSense where native mappings are not available.

Kibana Dashboards

Ingest Pipelines

- **Filebeat-style inputs** via Elastic Agent for eve.json, Apache logs, and custom JSON.
- **Syslog inputs** for pfSense and OPNSense firewall logs.
- **HTTP JSON inputs** for Authentik API event collection.
- **Processors** include:
 - `decode_json_fields` for NDJSON
 - `add_fields` for tagging source and dataset
 - `geoip` and `user_agent` enrichment where applicable

Dashboards

- **Suricata Overview:** Alerts by severity, protocol breakdown, top talkers.
- **Windows Sysmon:** Process tree visualizations, registry changes, execution anomalies.
- **Docker & Prometheus:** Container health, resource usage, service metrics.
- **Firewall Logs:** Blocked vs allowed traffic, source geolocation, port scans.
- **Authentik Events:** User logins, MFA challenges, admin actions.

All dashboards are built in Kibana using saved searches, visualizations, and Lens panels. Filters and drilldowns allow SOC-style triage and threat hunting.

Integrations

Choose an integration to start collecting and analyzing your data.

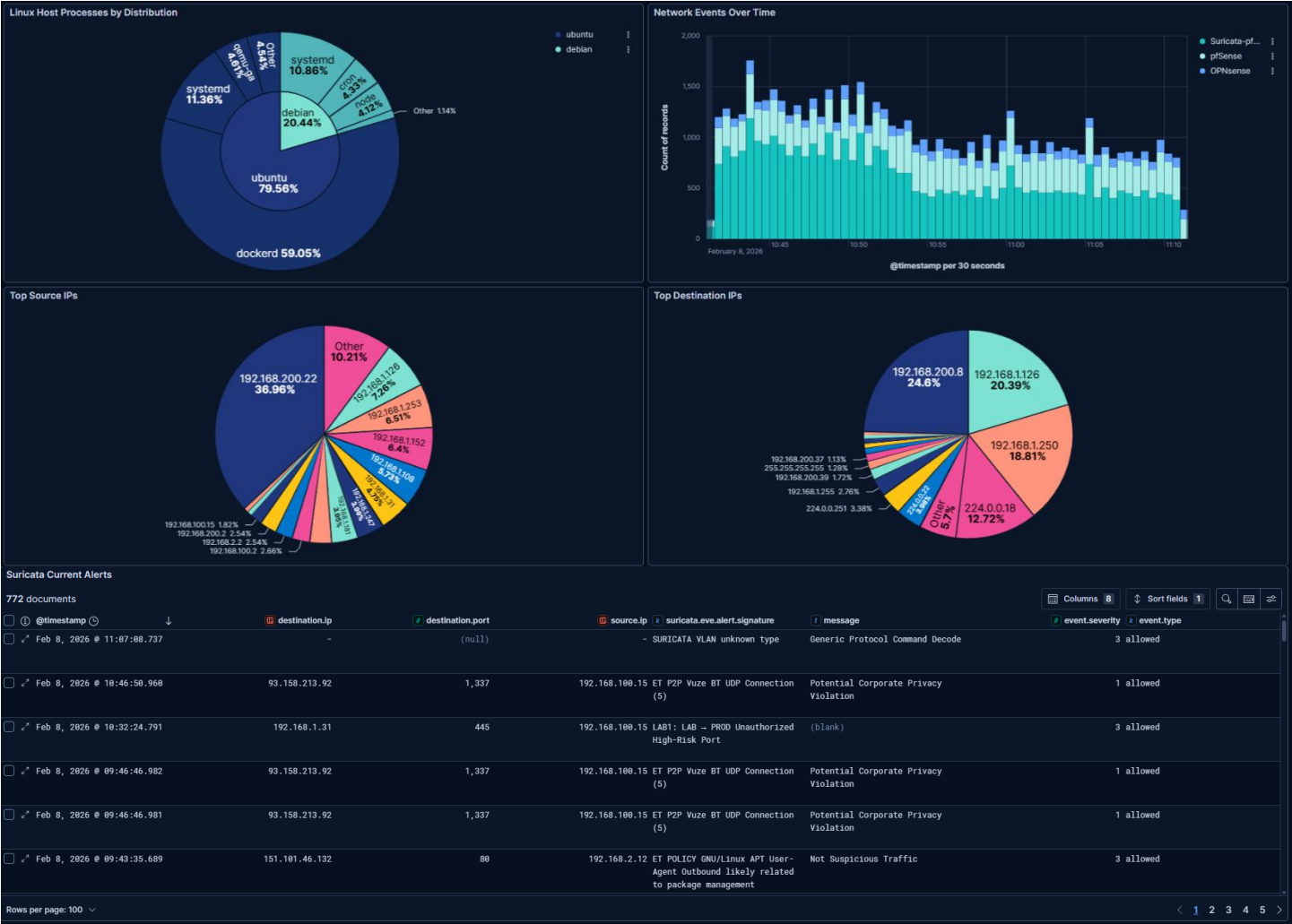
[Browse integrations](#)
[Installed integrations 1](#)

Search

Showing 20 integrations

☐	Integration name	Version
☐	Apache HTTP Server	2.1.1
☐	Auditd Logs	3.22.0
☐	authentik	1.6.1
☐	Cisco IOS	1.34.0
☐	Docker	2.14.0
☐	Elastic Agent	2.6.8
☐	Elastic Defend	9.2.0
☐	Fleet Server	1.6.0
☐	Kubernetes	1.81.1
☐	Network Packet Capture	1.34.0
☐	pfSense	1.24.0
☐	Prometheus	1.24.2
☐	Prebuilt Security Detection Rules	Upgrade
☐	Snort	1.20.0
☐	Suricata	2.26.0
☐	Elastic Synthetics	1.4.2
☐	System	2.7.2
☐	Threat Map	1.1.0
☐	Traefik	2.7.0
☐	Windows	3.2.3

Custom Lab Dashboard



Authentik Authentication Dashboard

Visualizations:

- Login Success Rate: Pie chart showing successful vs failed logins
- Login Attempts by User: Bar chart
- Geographic Login Distribution: Map of source.geo.location
- Authentication Timeline: Bar chart of login events
- Top Applications Accessed: Pie chart of event.dataset†



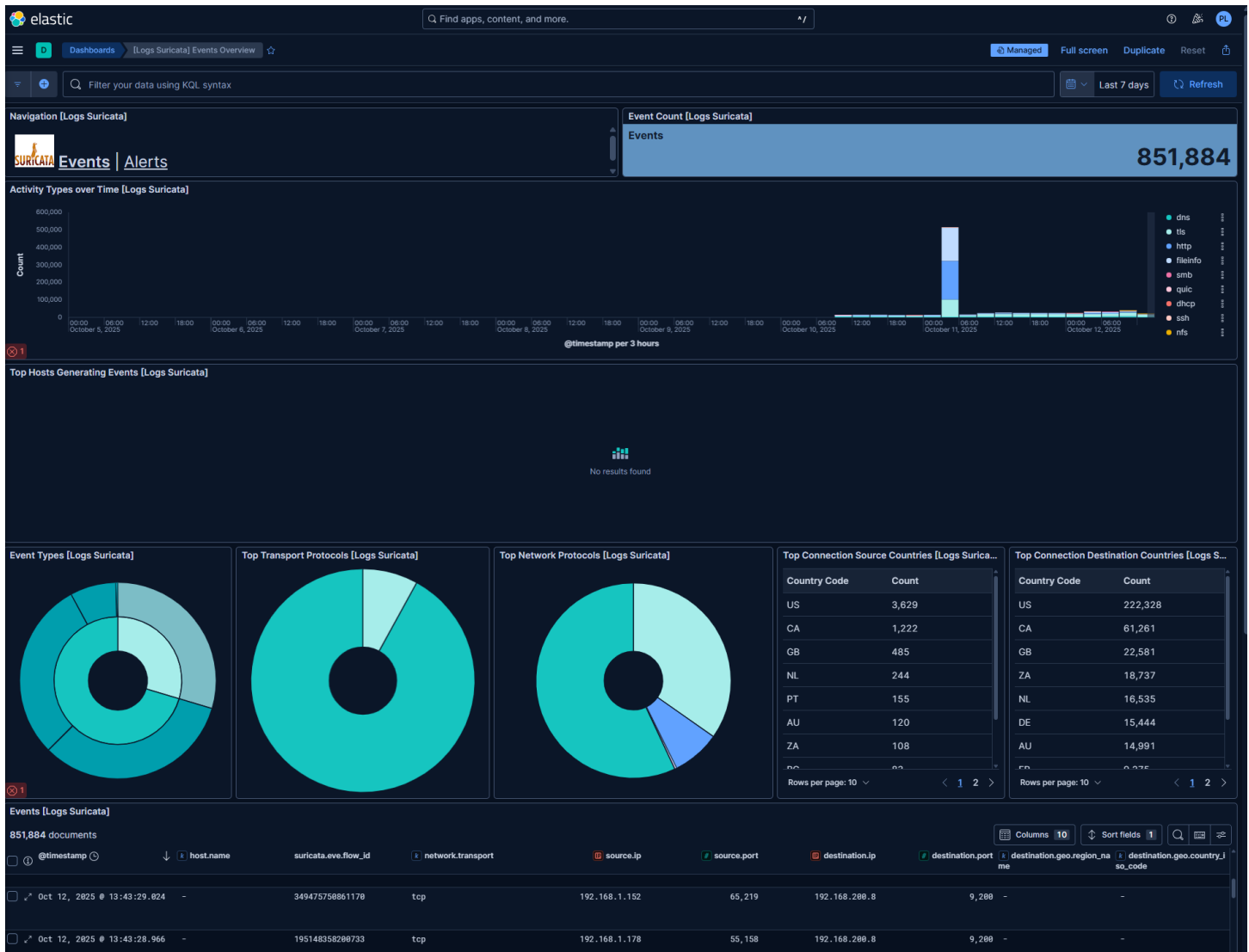
Suricata IDS Dashboard

Visualizations:

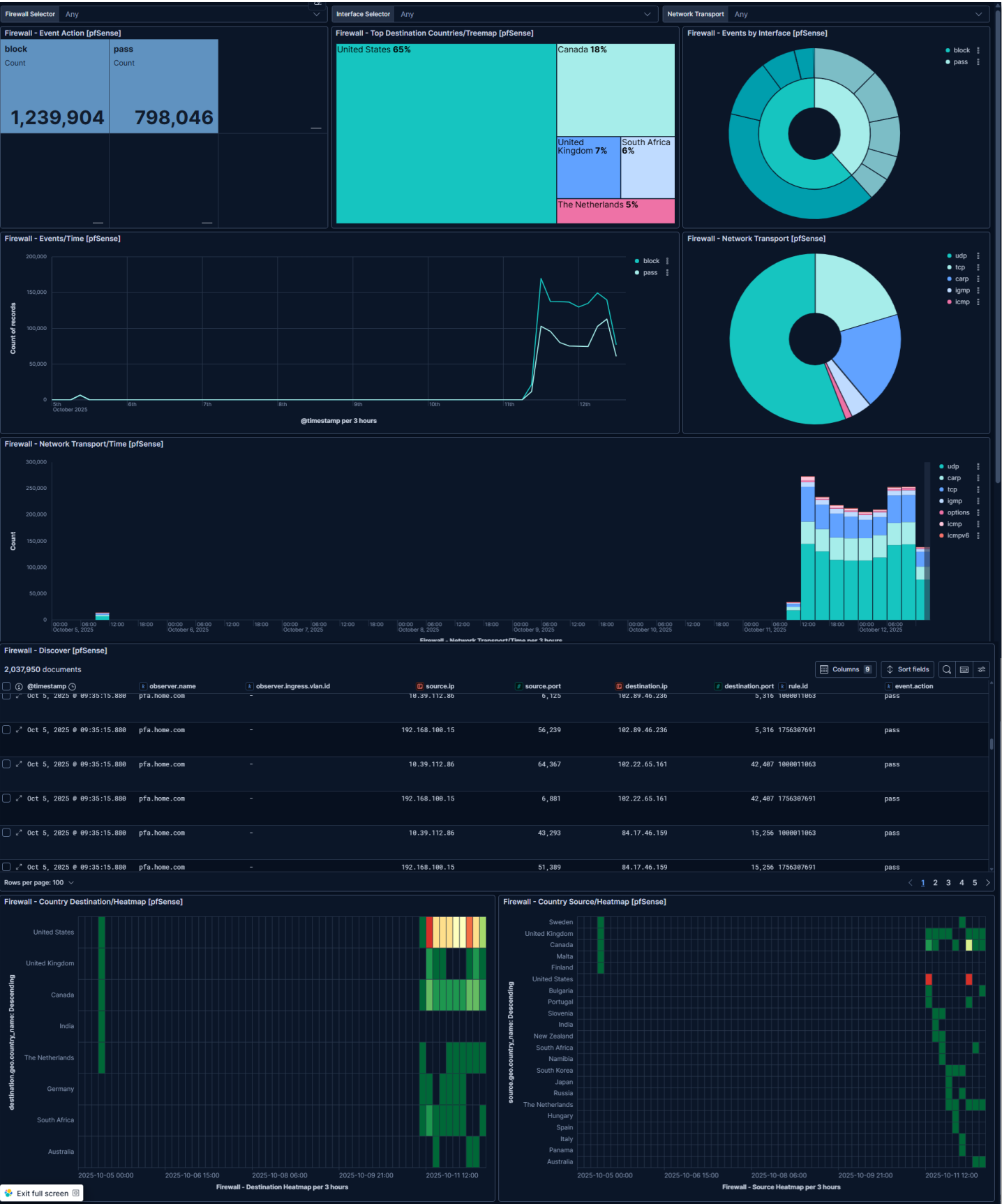
- Event Count
- Alert Severity Distribution: Pie chart (Critical, High, Medium, Low, Info)
- Activity types Over Time: Stacked bar chart with 5-minute buckets
- Top Attack Signatures: Data table with alert.signature and count
- Protocol Breakdown: Donut chart of network.protocol
- Destination Ports: Heatmap of frequently targeted ports

Example Kibana Query (KQL):

event.dataset: suricata.eve AND event.kind: alert AND alert.severity: [1 TO 2]

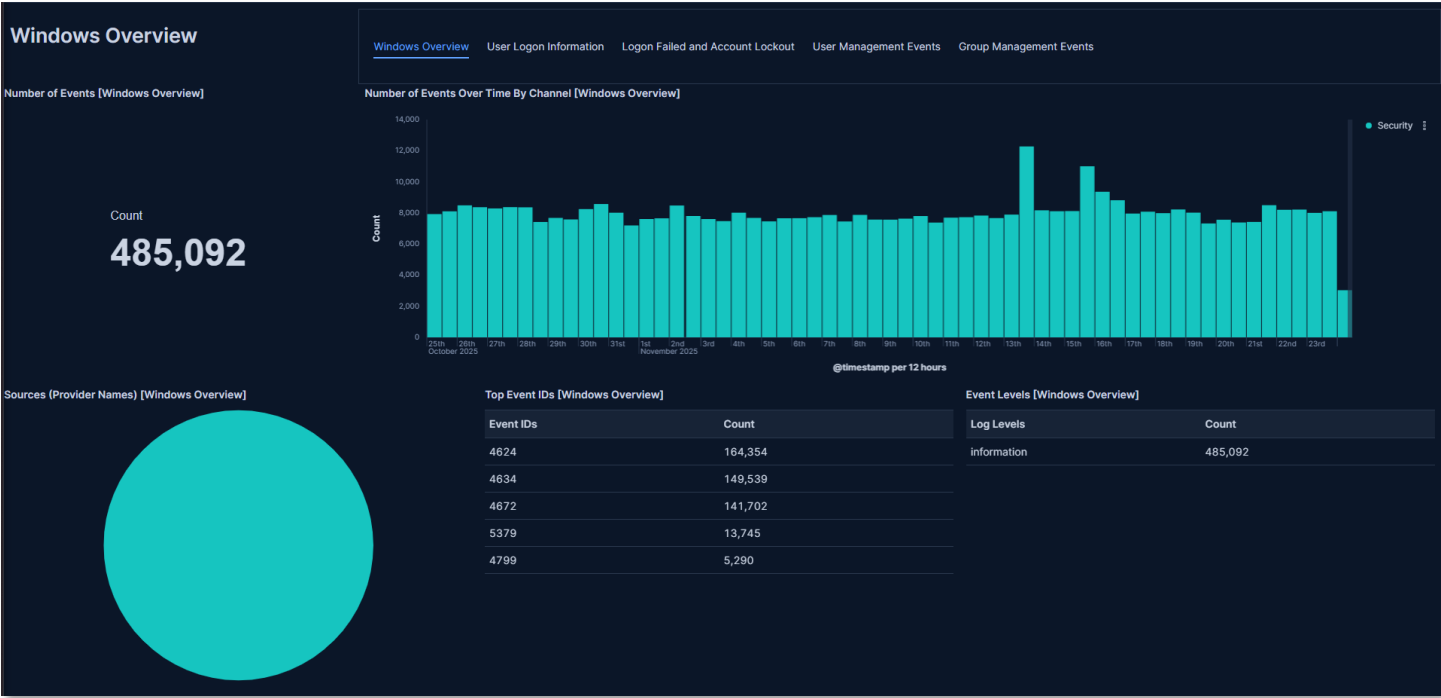


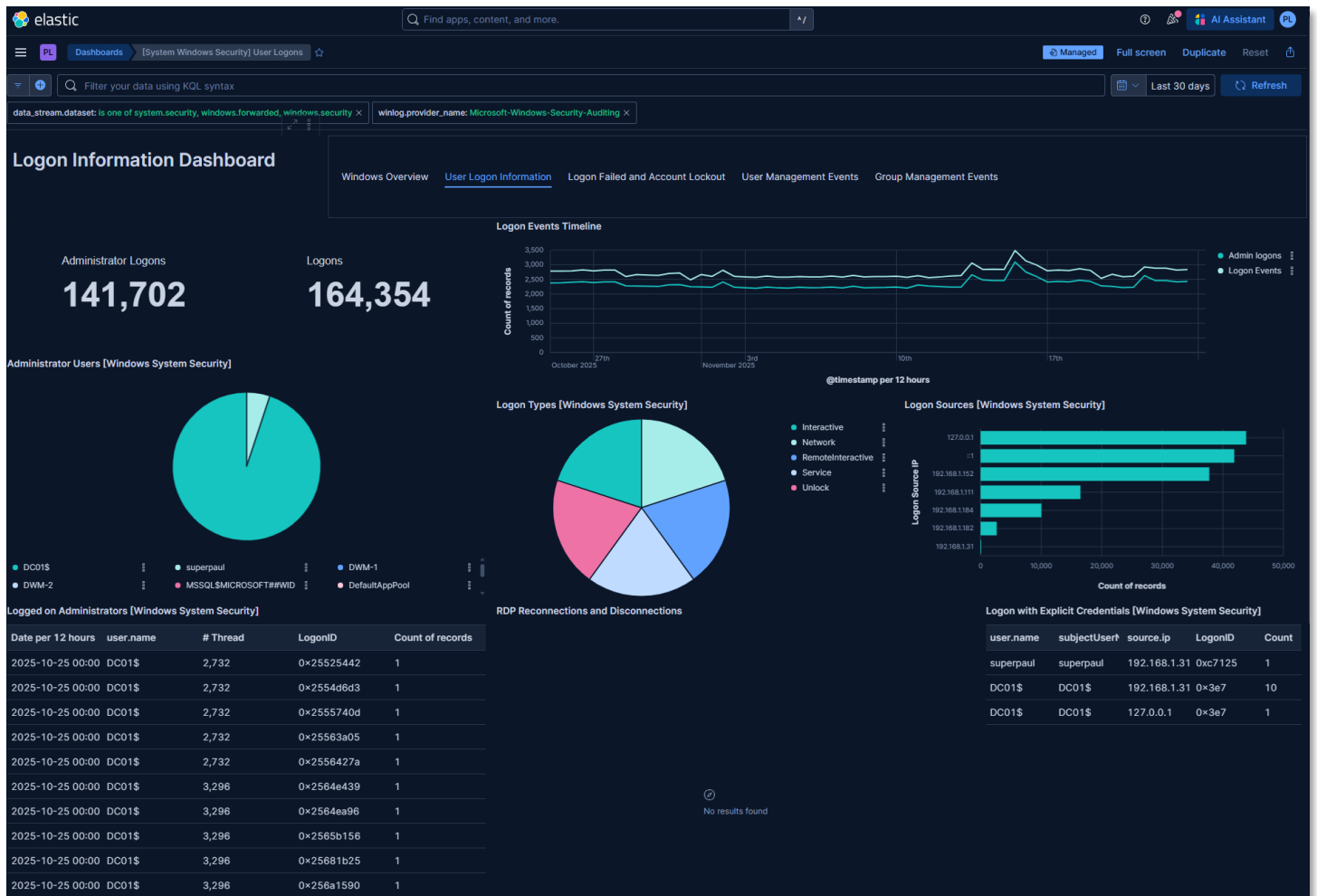
pfSense Firewall Filter Log and Location Dashboard



Windows Overview and Security Dashboards:
Visualizations:

- Number of events
- Top Event IDs
- Number of Events Over Time by Channel: Bar chart of events by date
- Total Administrator and standard logins
- Login Event Timeline: line graph of administrator and standard login events
- Administrator Users: Pie chart of logins by account
- Login Types: Pie chart of the login event by type
- Login Sources: Bar chart of the logins by source IP
- Login events: table of login event by account, source IP and threads

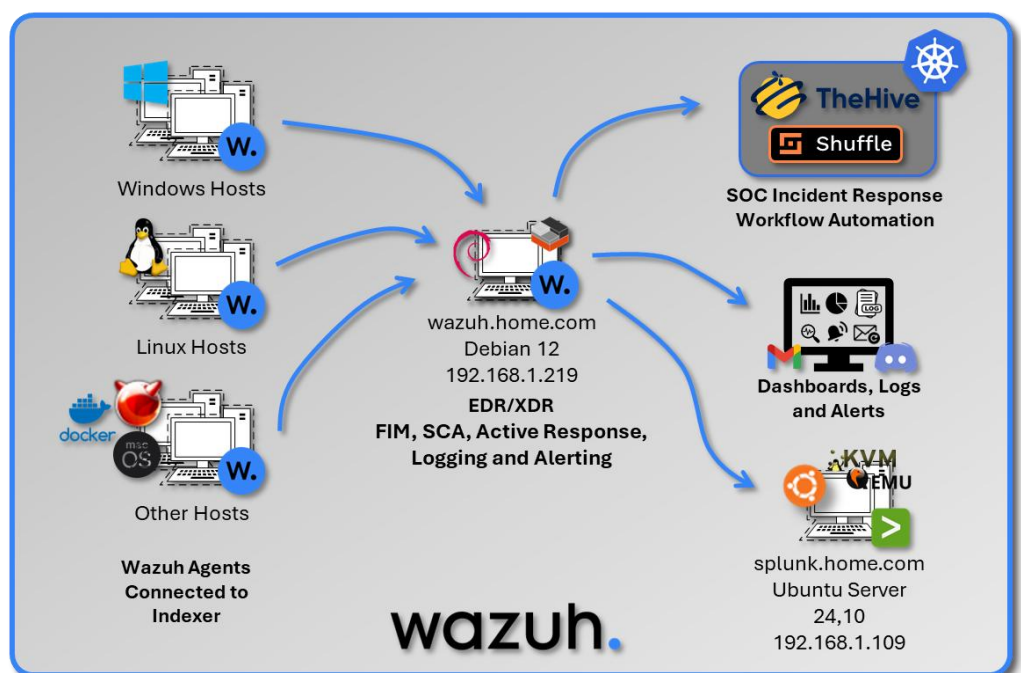




Wazuh Endpoint Detection and Response (EDR)

Deployment Overview

Wazuh provides host-based security monitoring, threat detection, and compliance auditing across Windows, Linux, BSD, and macOS endpoints. The platform combines SIEM-style log analysis with EDR-grade telemetry to deliver comprehensive endpoint visibility. Wazuh Agents collect file integrity data, vulnerability information, authentication logs, and system events, forwarding them to a centralized Wazuh Manager for correlation and alerting. The



deployment integrates with both Splunk and Elastic to unify endpoint telemetry with broader security analytics.

Security Impact

Wazuh strengthens endpoint security by detecting unauthorized file changes, rootkits, suspicious processes, and privilege escalation attempts. File Integrity Monitoring (FIM) provides early detection of tampering with critical system files, while rootkit detection identifies hidden processes and kernel-level threats. Vulnerability assessment scans endpoints for missing patches and known CVEs, and Security Configuration Assessment (SCA) enforces CIS benchmark compliance across multiple operating systems. Active Response automates containment actions such as firewall drops, host isolation, and account disabling. VirusTotal integration validates suspicious files, and Discord notifications provide real-time alerting. Comprehensive audit logs support forensic investigations and incident reconstruction.

Deployment Rationale

Endpoint compromise remains one of the most common initial access vectors in modern attacks. Wazuh provides essential host-level visibility and automated response capabilities that complement network-based and SIEM-based detection. Its multi-OS support ensures consistent monitoring across heterogeneous environments, while its integration with Splunk and Elastic enables centralized correlation and threat hunting. This deployment demonstrates core EDR competencies including FIM, rootkit detection, vulnerability assessment, compliance auditing, and automated containment.

Architecture Principles Alignment

Defense in Depth

- Adds endpoint-layer detection to complement network, identity, and application telemetry
- FIM, rootkit detection, and vulnerability scanning provide overlapping security controls
- Active Response automates containment to reduce dwell time and limit lateral movement

Secure by Design

- CIS benchmark audits enforce hardened configurations across Windows, Linux, and application stacks
- SCA policies detect deviations from approved security baselines
- VirusTotal integration validates suspicious artifacts before quarantine or remediation

Zero Trust

- No endpoint activity is implicitly trusted; all events are logged, analyzed, and correlated
- Continuous monitoring ensures every process, file change, and authentication event is observable
- Automated containment actions enforce least-privilege and rapid isolation of compromised hosts

Wazuh agents are deployed across multiple endpoints to provide real-time monitoring, threat detection, and compliance auditing.

Each agent reports to the Wazuh Manager (node01) and is monitored via the Wazuh dashboard for:

- File integrity monitoring
- Rootkit detection
- Vulnerability assessment
- Compliance reporting (e.g., HIPAA, NIST)

Component Architecture

Deployment Model: Single-node manager running on an LXC, distributed agents

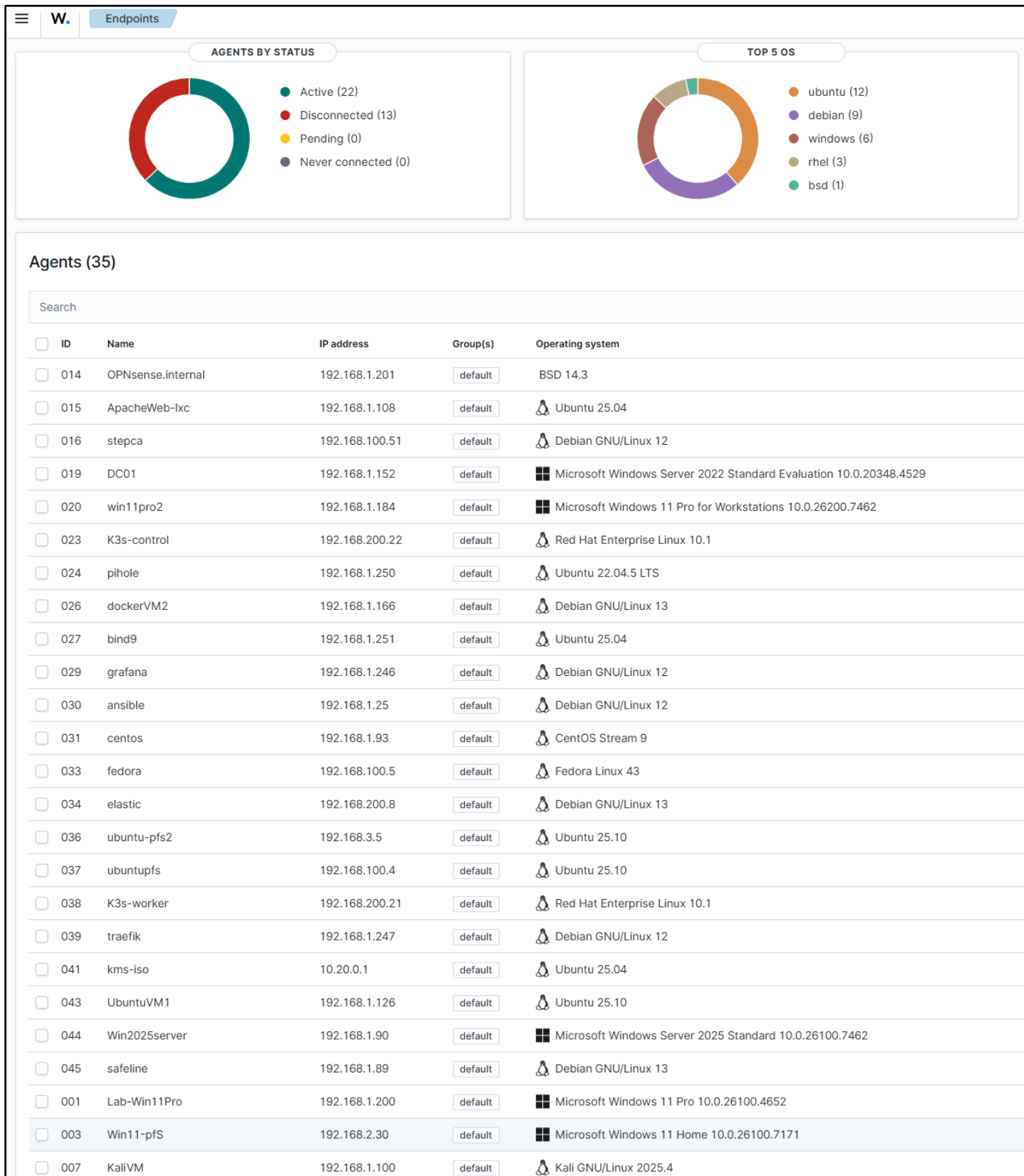
Components:

- Wazuh Manager – Core analysis and correlation engine

- Wazuh Indexer – Stores and indexes security events
- Wazuh Dashboard – Web UI for search, visualization, and alert management
- Agents – Installed on monitored endpoints
- Splunk Universal Forwarder – log export for systemwide normalization and correlation

Agent List:

ID	Name	IP address	OS Platform and Release
0	wazuh	127.0.0.1	Debian GNU/Linux, 12(bookworm)
1	Lab-Win11Pro	192.168.1.200	Microsoft Windows 11 Pro, 10.0.26100.4652(N/A)
3	Win11-pfS	192.168.2.30	Microsoft Windows 11 Home, 10.0.26100.7171(N/A)
7	KaliVM	192.168.1.100	Kali GNU/Linux, 2025.4(N/A)
13	Logans-MacBook-Air.local	192.168.1.207	macOS, 14.7.8(Sonoma)
14	OPNsense.internal	192.168.1.201	FreeBSD, 14.3(N/A)
15	ApacheWeb-lxc	192.168.1.108	Ubuntu, 25.04(Plucky Puffin)
16	stepca	192.168.100.51	Debian GNU/Linux, 12(bookworm)
17	Win11Pro	192.168.1.111	Microsoft Windows 11 Pro, 10.0.26100.7462(N/A)
18	UbuntuGenVM	192.168.1.126	Ubuntu, 25.1(Questing Quokka)
19	DC01	192.168.1.152	Microsoft Windows Server 2022 Standard Evaluation, 10.0.20348.4529(N/A)
20	win11pro2	192.168.1.184	Microsoft Windows 11 Pro for Workstations, 10.0.26200.7462(N/A)
21	Plex-lxc	192.168.1.136	Ubuntu, 22.04.5 LTS(Jammy Jellyfish)
22	redhat-control	192.168.200.21	Red Hat Enterprise Linux, 10(Coughlan)
23	K3s-control	192.168.200.22	Red Hat Enterprise Linux, 10.1(Coughlan)
24	Technetium-dns01	192.168.1.150	Debian GNU/Linux, 13(trixie)
26	dockerVM2	192.168.1.166	Debian GNU/Linux, 13(trixie)
27	Technetium-dns02	192.168.1.151	Debian GNU/Linux, 13(trixie)
28	Unbound-01	192.168.1.153	Debian GNU/Linux, 13(trixie)
29	grafana	192.168.1.246	Debian GNU/Linux, 12(bookworm)
30	ansible	192.168.1.25	Debian GNU/Linux, 12(bookworm)
31	centos	192.168.1.93	CentOS Stream, 9(N/A)
32	splunk	192.168.1.109	Ubuntu, 24.1(Oracular Oriole)
33	fedora	192.168.100.5	Fedora Linux, 43(Server Edition)
34	elastic	192.168.200.8	Debian GNU/Linux, 13(trixie)
35	Ubuntu-pfLAN	192.168.2.7	Ubuntu, 25.1(Questing Quokka)
36	ubuntu-pfs2	192.168.3.5	Ubuntu, 25.1(Questing Quokka)
37	ubuntupfs	192.168.100.4	Ubuntu, 25.1(Questing Quokka)
38	K3s-worker	192.168.200.21	Red Hat Enterprise Linux, 10.1(Coughlan)
39	traefik	192.168.1.247	Debian GNU/Linux, 12(bookworm)
40	debian-Extlan	192.168.2.5	Debian GNU/Linux, 12(bookworm)
41	kms-iso	10.20.0.1	Ubuntu, 25.04(Plucky Puffin)
42	parrotOS	192.168.100.16	Parrot Security, 7.1(echo)
43	UbuntuVM1	192.168.1.126	Ubuntu, 25.1(Questing Quokka)
44	Win2025server	192.168.1.90	Microsoft Windows Server 2025 Standard, 10.0.26100.7462(N/A)
45	safeline	192.168.1.89	Debian GNU/Linux, 13(trixie)



Security Monitoring Modules:

File Integrity Monitoring (FIM):

Purpose: Detect unauthorized changes to critical system files and attacker-target directories.

Configuration:

- Real-time monitoring: Enabled for critical paths and high-risk directories
- Linux: /root, /tmp, /var/tmp, /home/*/Downloads
- Windows: C:\Users*\Downloads, C:\Windows\Temp, C:\Temp, C:\Users*\AppData\Local\Temp
- Scheduled scans: Every 12 hours for full filesystem baseline
- Hash algorithms: MD5, SHA1, SHA256
- Attributes tracked: Size, permissions, ownership, mtime
- Alert on: Creation, modification, deletion

Example FIM Alerts:

Description: New file created in /tmp

File: /tmp/malicious.sh

Action: Alert + snapshot of file metadata

Description: Integrity checksum changed

File:

C:\Windows\System32\drivers\etc\hosts

Old MD5: abc123...

New MD5: def456...

Action: Alert + snapshot of file content

Rootkit Detection:

Purpose: Identify hidden processes, ports, and kernel modules

```
<!-- File Integrity Monitoring -->
<syscheck>
  <disabled>no</disabled>
  <frequency>43200</frequency>
  <scan_on_start>yes</scan_on_start>
  <alert_new_files>yes</alert_new_files>
  <auto_ignore frequency="10" timeframe="3600">no</auto_ignore>

  <!-- Critical system dirs -->
  <directories realtime="yes">/etc</directories>
  <directories realtime="yes">/usr/bin</directories>
  <directories realtime="yes">/usr/sbin</directories>
  <directories realtime="yes">/bin</directories>
  <directories realtime="yes">/sbin</directories>
  <directories realtime="yes">/boot</directories>

  <!-- Threat actor targets -->
  <directories realtime="yes">/root</directories>
  <directories realtime="yes">/tmp</directories>
  <directories realtime="yes">/var/tmp</directories>
  <directories realtime="yes">/home/*/Downloads</directories>
  <directories realtime="yes">C:\Users\*\Downloads</directories>
  <directories realtime="yes">C:\Windows\Temp</directories>
  <directories realtime="yes">C:\Temp</directories>
  <directories realtime="yes">C:\Users\*\AppData\Local\Temp</directories>
```

Checks Performed:

- Hidden processes: Compare ps output with /proc
- Hidden ports: Netstat vs /proc/net/* comparison
- Kernel modules: Verify against known-good list
- System binaries: Check for trojaned ls, ps, netstat
- Registry anomalies: Windows registry rootkits

Vulnerability Assessment:

Purpose: Identify missing patches and known CVEs

Vulnerability Sources:

- Windows: Microsoft Security Update Guide (MSU)
- Red Hat/CentOS: OVAL feeds
- Debian/Ubuntu: Debian Security Tracker
- macOS: Apple Security Updates

Example Vulnerability Alert:

CVE: CVE-2024-43451

Title: NTLM Hash Disclosure Vulnerability
CVSS: 8.1 (High)

Affected: Windows Server 2022 DC01

Package: ntlm.dll version 10.0.20348.2227

Fix Available: Yes (KB5043936)

Recommendation: Apply security update immediately

Security Configuration Assessment (SCA):

Purpose: Audit compliance with security benchmarks

Policies Applied:

- CIS Windows Server 2022 Benchmark
- CIS Windows 11 Benchmark
- CIS Debian 12 Benchmark
- CIS Debian 13 Benchmark
- CIS Ubuntu 22.04 LTS Benchmark
- CIS Red Hat Enterprise Linux 10 Benchmark
- CIS CentOS Linux 9 Benchmark
- CIS Distribution Independent Linux Benchmark v2.0.0
- CIS Apache 2.4 Benchmark
- PCI-DSS v4.0 requirements
- NIST 800-53 controls

Example SCA Finding:

Policy: CIS Windows Server 2022

```
<!-- Rootcheck (policy monitoring) -->
<rootcheck>
  <disabled>no</disabled>
  <check_files>yes</check_files>
  <check_trojans>yes</check_trojans>
  <check_dev>yes</check_dev>
  <check_sys>yes</check_sys>
  <check_pids>yes</check_pids>
  <check_ports>yes</check_ports>
  <check_if>yes</check_if>
  <frequency>43200</frequency>
  <rootkit_files>etc/rootcheck/rootkit_files.txt</rootkit_files>
  <rootkit_trojans>etc/rootcheck/rootkit_trojans.txt</rootkit_trojans>
  <skip_nfs>yes</skip_nfs>
  <ignore>/var/lib/containerd</ignore>
  <ignore>/var/lib/docker/overlay2</ignore>
</rootcheck>

<!-- System inventory -->
<wodle name="syscollector">
  <disabled>no</disabled>
  <interval>1h</interval>
  <scan_on_start>yes</scan_on_start>
  <hardware>yes</hardware>
  <os>yes</os>
  <network>yes</network>
  <packages>yes</packages>
  <ports all="no">yes</ports>
  <processes>yes</processes>
  <synchronization>
    <max_eps>10</max_eps>
  </synchronization>
</wodle>

<!-- Security Configuration Assessment -->
<sca>
  <enabled>yes</enabled>
  <scan_on_start>yes</scan_on_start>
  <interval>12h</interval>
  <skip_nfs>yes</skip_nfs>
</sca>

<!-- Vulnerability detection -->
<vulnerability-detection>
  <enabled>yes</enabled>
  <index-status>yes</index-status>
  <feed-update-interval>60m</feed-update-interval>
</vulnerability-detection>
```

Check ID: 2.3.10.5

Title: Network access: Do not allow storage of passwords and credentials

Status: FAIL

Current Value: Enabled

Expected Value: Disabled

Remediation: Set HKLM\SYSTEM\CurrentControlSet\Control\Lsa\DisableDomainCreds to 1

Severity: High

Login Monitoring (SSH/RDP)

Purpose: Detect brute force attempts and unauthorized remote access.

Configuration:

- Linux: Monitor /var/log/auth.log for failed SSH login attempts
 - Windows: Monitor Security event channel for Event ID 4625 (failed RDP logins)
 - Thresholds: Multiple failed attempts within a short timeframe trigger high-severity alerts
- Example Alert:
- Rule ID: 5710
 - Description: Multiple failed SSH login attempts detected
 - Source IP: 192.168.3.45
 - Action: Alert + optional active response (firewall drop)

Active Response

Firewall Drop

Command:

```
<command>
  <name>firewall-drop</name>
  <executable>firewall-drop</executable>
  <expect>srcip</expect>
  <timeout_allowed>yes</timeout_allowed>
</command>
```

Active Response:

```
<active-response>
  <command>firewall-drop</command>
  <location>any</location>
  <rules_group>authentication_failed,sshd
</rules_group>
  <timeout>600</timeout>
</active-response>
```

```
<!-- Active response -->
<active-response>
  <command>discord</command>
  <location>local</location>
  <rules_group>sshd,| authentication_failure,| syscheck_file</rules_group>
</active-response>

<active-response>
  <disabled>no</disabled>
  <command>remove-threat</command>
  <location>local</location>
  <rules_id>87105</rules_id>
</active-response>

<active-response>
  <command>firewall-drop</command>
  <location>any</location>
  <rules_group>authentication_failed,sshd</rules_group>
  <timeout>600</timeout>
</active-response>

<active-response>
  <command>host-deny</command>
  <location>any</location>
  <rules_group>authentication_failed</rules_group>
  <timeout>600</timeout>
</active-response>

<active-response>
  <command>disable-account</command>
  <location>local</location>
  <rules_group>authentication_failed</rules_group>
</active-response>
```

- **Alerts/Trigger:** Fires when rules in the authentication_failed or sshd groups are triggered (e.g., repeated failed SSH logins).
- **Workflow:** Wazuh passes the offending source IP (srcip) to the firewall-drop script, which inserts an iptables rule to block traffic. After 600 seconds, the block is lifted.
- **Benefit:** Immediate containment of brute-force attackers, reducing exposure while still allowing legitimate traffic after timeout.

Host Deny

Command

```
<command>
```

```
<name>host-deny</name>
<executable>host-deny</executable>
<expect>srcip</expect>
<timeout_allowed>yes</timeout_allowed>
</command>
```

Active Response

```
<active-response>
  <command>host-deny</command>
  <location>any</location>
  <rules_group>authentication_failed</rules_group>
  <timeout>600</timeout>
</active-response>
```

- **Alerts/Trigger:** Activated on authentication_failed events (e.g., failed login attempts).
- **Workflow:** The offending IP is added to /etc/hosts.deny, preventing further connections. After 600 seconds, the entry is removed.
- **Benefit:** Lightweight containment using TCP wrappers; useful on systems where iptables isn't preferred or available.

Disable Account

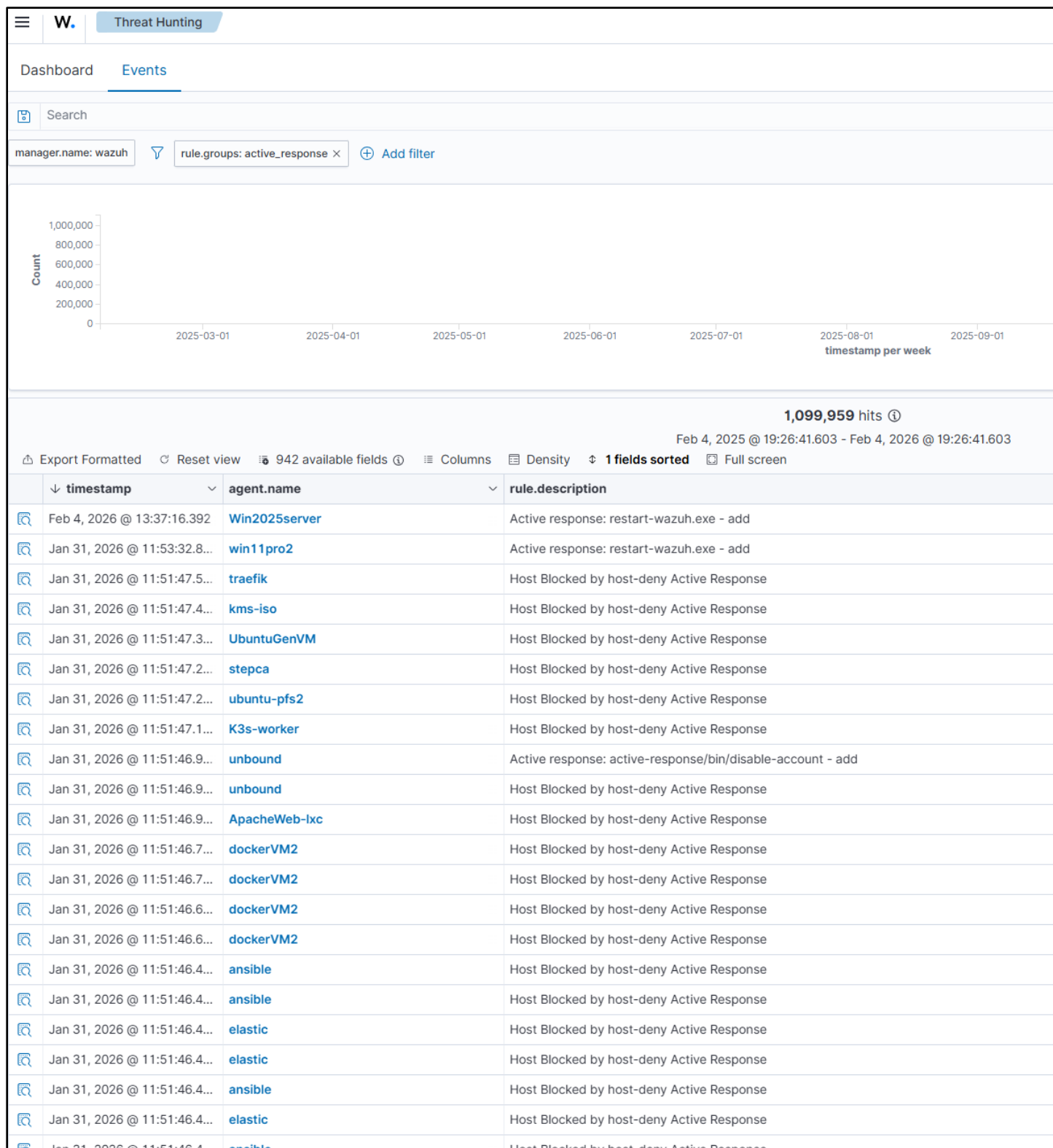
Command

```
<command>
  <name>disable-account</name>
  <executable>disable-account</executable>
  <expect>user</expect>
  <timeout_allowed>no</timeout_allowed>
</command>
```

Active Response

```
<active-response>
  <command>disable-account</command>
  <location>local</location>
  <rules_group>authentication_failed</rules_group>
</active-response>
```

- **Alerts/Trigger:** Fires when a local user repeatedly fails authentication (authentication_failed).
- **Workflow:** Wazuh passes the username to the disable-account script, which locks the account immediately. No timeout is allowed — the account stays disabled until an admin intervenes.
- **Benefit:** Prevents compromised or brute-forced accounts from being abused. Strong containment at the user level, especially for privileged accounts.



Integrations

Splunk

Log Forwarding Configuration:

Wazuh Manager → Splunk Universal Forwarder → Splunk Indexer

Forwarder Configuration (inputs.conf):

```
[monitor:///var/ossec/logs/alerts/alerts.json]
disabled = false
sourcetype = wazuh:alerts
index = wazuh-alerts
source = wazuh-manager
```

Alert Enrichment in Splunk:

- MITRE ATT&CK Mapping: Techniques and tactics
- Asset Context: Host criticality, owner, location
- Threat Intelligence: IOC matching from feeds
- User Context: AD user attributes, group membership

VirusTotal

VirusTotal is integrated into the File Integrity Monitoring workflow through a custom rule that targets new or modified executable and script files. When rule_id 87105 is triggered — indicating VirusTotal has flagged a file with multiple positive detections — a host-level Bash script automatically quarantines the file for remediation. This ensures only high-risk binaries are acted upon, while preserving quota and reducing false positives.

- Scope
 - Applies to File Integrity Monitoring events involving executable and script files in monitored directories.
 - Focused only on high-risk file types to conserve VirusTotal API quota.
- Trigger
 - Custom rule_id 87105 fires when VirusTotal reports multiple positive detections for a monitored file.
 - Detection is based on FIM alerts enriched with VirusTotal validation.
- Action
 - A host level Bash script is invoked automatically.
 - The script quarantines or removes the flagged file, preventing execution or further spread.
 - Response is logged for auditability and traceability.
- Benefit
 - Ensures only suspicious binaries are escalated, reducing false positives.
 - Preserves VirusTotal quota by filtering out irrelevant file changes.
 - Provides automated containment, minimizing manual intervention.
 - Enhances security posture by integrating detection, validation, and remediation into a single workflow.

```
<integration>
  <name>virustotal</name>
  <api_key>xxxxxxxxxxxxxxxxxxxx</api_key>
  <rule_id>100003</rule_id>
  <alert_format>json</alert_format>
</integration>
```

```
<!-- Custom VirusTotal rule -->
<group name="local,syscheck,virustotal,">

  <!--
  Trigger only when syscheck detects changes to executable or script files
  -->

  <rule id="100003" level="10">
    <if_group>syscheck</if_group>
    <match>\.(exe|dll|sh|py|bin|pl|jar)$</match>
    <description>Syscheck: suspicious file change (exec/script) sent to VirusTotal</description>
    <group>virustotal,syscheck_file,</group>
  </rule>
```

```
<command>
  <name>remove-threat</name>
  <executable>remove-threat.sh</executable>
  <timeout_allowed>no</timeout_allowed>
</command>
```

```
<active-response>
  <disabled>no
</disabled>
```

```
<command>remove-  
threat</command>  
  <location>local  
</location>  
  <rules_id>  
87105</rules_id >Discord  
</active-response>
```

```

1  #!/bin/bash
2
3  LOCAL="dirname $0";
4  cd $LOCAL
5  cd ../
6
7  PWD=`pwd`
8
9  read INPUT_JSON
10 FILENAME=$(echo $INPUT_JSON | jq -r .parameters.alert.data.virustotal.source.file)
11 COMMAND=$(echo $INPUT_JSON | jq -r .command)
12 LOG_FILE=$(PWD)/../logs/active-responses.log"
13
14 #----- Analyze command -----#
15 if [ ${COMMAND} = "add" ]
16 then
17     # Send control message to execd
18     printf '{"version":1,"origin":{"name":"remove-threat","module":"active-response"},"command":"check_keys", "parameters":{"keys":[]}}\n'
19
20     read RESPONSE
21     COMMAND2=$(echo $RESPONSE | jq -r .command)
22     if [ ${COMMAND2} != "continue" ]
23     then
24         echo ""date '+%Y/%m/%d %H:%M:%S'" $0: $INPUT_JSON Remove threat active response aborted" >> ${LOG_FILE}
25         exit 0;
26     fi
27 fi
28
29 # Removing file
30 rm -f $FILENAME
31 if [ $? -eq 0 ]; then
32     echo ""date '+%Y/%m/%d %H:%M:%S'" $0: $INPUT_JSON Successfully removed threat" >> ${LOG_FILE}
33 else
34     echo ""date '+%Y/%m/%d %H:%M:%S'" $0: $INPUT_JSON Error removing threat" >> ${LOG_FILE}
35 fi
36
37 exit 0;

```

▼ syscheck.path

```
/tmp/malware/eicar
```

```
/tmp/malware/eicar
```

timestamp	agent.name	syscheck.path	syscheck.event	rule.description	rule.level	rule.id
Nov 27, 2025 @ 08:53:27.0...	ApacheWeb-ixc	/tmp/malware/eicar	added	File added to the system.	5	554
Nov 27, 2025 @ 08:53:19.2...	ApacheWeb-ixc	/tmp/malware/eicar	deleted	File deleted.	7	553

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651ddf

🔍

📄

File distributed by Offensive Security

🔄 Reanalyze ⌵ Similar ⌵ More ⌵

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651ddf

eicar.com-8491

Size
68 B

Last Analysis Date
4 minutes ago

📄

🔗

powershell long-sleeps detect-debug-environment idle known-distributor direct-cpu-clock-access via-tor attachment

Discord

External alerting discord has been configured for specific workflows. Alerts from the sshd, authentication_failure or syscheck_file groups trigger a custom script that sends a message to a Discord webhook in the #wazuh channel. Alerts are enriched with rule, group and host information for easy identification.

- Scope: Alerts generated by the following Wazuh rule groups:
 - sshd (SSH login events)
 - authentication_failure (failed login attempts)
 - syscheck_file (file integrity monitoring changes)
- Trigger: When any of these groups fire, Wazuh's active response mechanism invokes a custom Bash script.
- Action:
 - The script formats the alert into a Discord embed.
 - It enriches the message with key metadata:
 - Rule → description of the triggered rule
 - Group → the rule group classification
 - Host → the agent name or hostname where the event occurred
 - Delivery: The enriched alert is sent to a Discord webhook configured for the #wazuh channel.
- Benefit:
 - Provides real time visibility into critical security events.
 - Alerts are actionable and easy to identify thanks to rule, group, and host context.
 - Centralizes notifications in Discord, enabling faster collaboration and response.



```
<integration>
  <name>discord</name>
  <hook_url>
https://discord.com/api/webhooks/xxxxxxx
  </hook_url>
  <group>multiple_drops,
authentication_failed,sshd,syscheck_file
  </group>
  <alert_format>json</alert_format>
</integration>

<command>
  <name>discord</name>
  <executable>discord.sh</executable>
  <expect>json</expect>
  <timeout_allowed>no</timeout_allowed>
</command>

<active-response>
  <disabled>no</disabled>
  <command>discord</command>
```

```
Wazuh > $ discord.sh > ...
1  #!/bin/bash
2  WEBHOOK_URL="https://discord.com/api/webhooks/1418221255097847818/ip7ue9
3
4  # Arguments passed from Wazuh active response
5  ALERT_JSON=$(cat)
6
7  # Extract useful fields from the alert JSON
8  RULE_NAME=$(echo "$ALERT_JSON" | jq -r '.rule.description')
9  RULE_GROUP=$(echo "$ALERT_JSON" | jq -r '.rule.groups[0]')
10 HOSTNAME=$(echo "$ALERT_JSON" | jq -r '.agent.name')
11
12 # Build Discord webhook payload
13
14 PAYLOAD=$(jq -n \
15   --arg rule "$RULE_NAME" \
16   --arg group "$RULE_GROUP" \
17   --arg host "$HOSTNAME" \
18   '{
19     "username": "Wazuh Alert",
20     "embeds": [
21       {
22         "title": "🚨 Wazuh Active Response Triggered",
23         "fields": [
24           {"name": "Rule", "value": $rule, "inline": true},
25           {"name": "Group", "value": $group, "inline": true},
26           {"name": "Host", "value": $host, "inline": true}
27         ],
28         "color": 15158332
29       }
30     ]
31   }')
32
33 # Send to Discord
34 curl -H "Content-Type: application/json" -d "$PAYLOAD" "$WEBHOOK_URL"
```

```
<location>local</location>
<rules_group>sshd,| authentication_failure,| syscheck_file</rules_group>
</active-response>
```

Shuffle SOAR

External alerting to Shuffle SOAR has been configured to automate case creation, enrichment, and downstream workflows. Alerts meeting the defined severity threshold are forwarded to a Shuffle webhook, where a workflow processes and routes the event for further action. Alerts are enriched with rule, agent, and metadata fields before being sent to TheHive or other SOAR components.

- Scope: Alerts generated by Wazuh with **severity ≥ 4** are forwarded to Shuffle for automated processing. This includes, but is not limited to, events from:
 - sshd (SSH authentication activity)
 - authentication_failure (failed login attempts)
 - syscheck_file (file integrity monitoring changes)
 - Any rule group producing level 4+ alerts
- Trigger: When an alert meets the configured severity threshold, Wazuh's integration engine sends the full JSON alert payload to the Shuffle webhook.
- Action: Inside Shuffle, the workflow performs the following steps:
 - Parses the incoming Wazuh alert JSON
 - Extracts key metadata fields:
 - **Rule** → rule description and rule ID
 - **Group** → Wazuh rule group classification
 - **Host** → agent name or hostname
 - **Severity** → numeric Wazuh level
 - Normalizes and sanitizes fields for TheHive compatibility
 - Creates a **TheHive Alert** using the enriched data
 - Routes alerts to additional automations such as:
 - Case Creation
 - Discord/email notifications
 - Cortex analyzers
- Delivery: The enriched alert is delivered to Shuffle via a webhook configured in Wazuh:

```
<integration>
  <name>shuffle</name>
  <hook_url>https://192.168.200.41/api/v1/hooks/webhook_32c000c1-3d4d-47f2-94d2-
f1c76c2f8c35</hook_url>
  <level>4</level>
  <alert_format>json</alert_format>
</integration>
```

 - Shuffle receives the alert, processes it through the workflow, and forwards the resulting alert object to TheHive for case management.
- Benefit:
 - Provides automated case creation for actionable alerts
 - Ensures consistent enrichment across all incoming events
 - Reduces manual triage time by routing alerts directly into TheHive
 - Enables multi-tool orchestration (Discord, Cortex, ticketing, containment)
 - Centralizes SOC automation in a single, extensible workflow engine

Alerting and Notifications

Purpose: Provide immediate visibility into critical events.

Configuration:

- Email Alerts (Outlook):
 - Wazuh configured to send alerts via local relay (msmtp/Postfix) → smtp.office365.com:587
 - Alerts delivered to recipient@outlook.com
- Discord Alerts:
 - Custom active response script posts alerts to Discord webhook channel

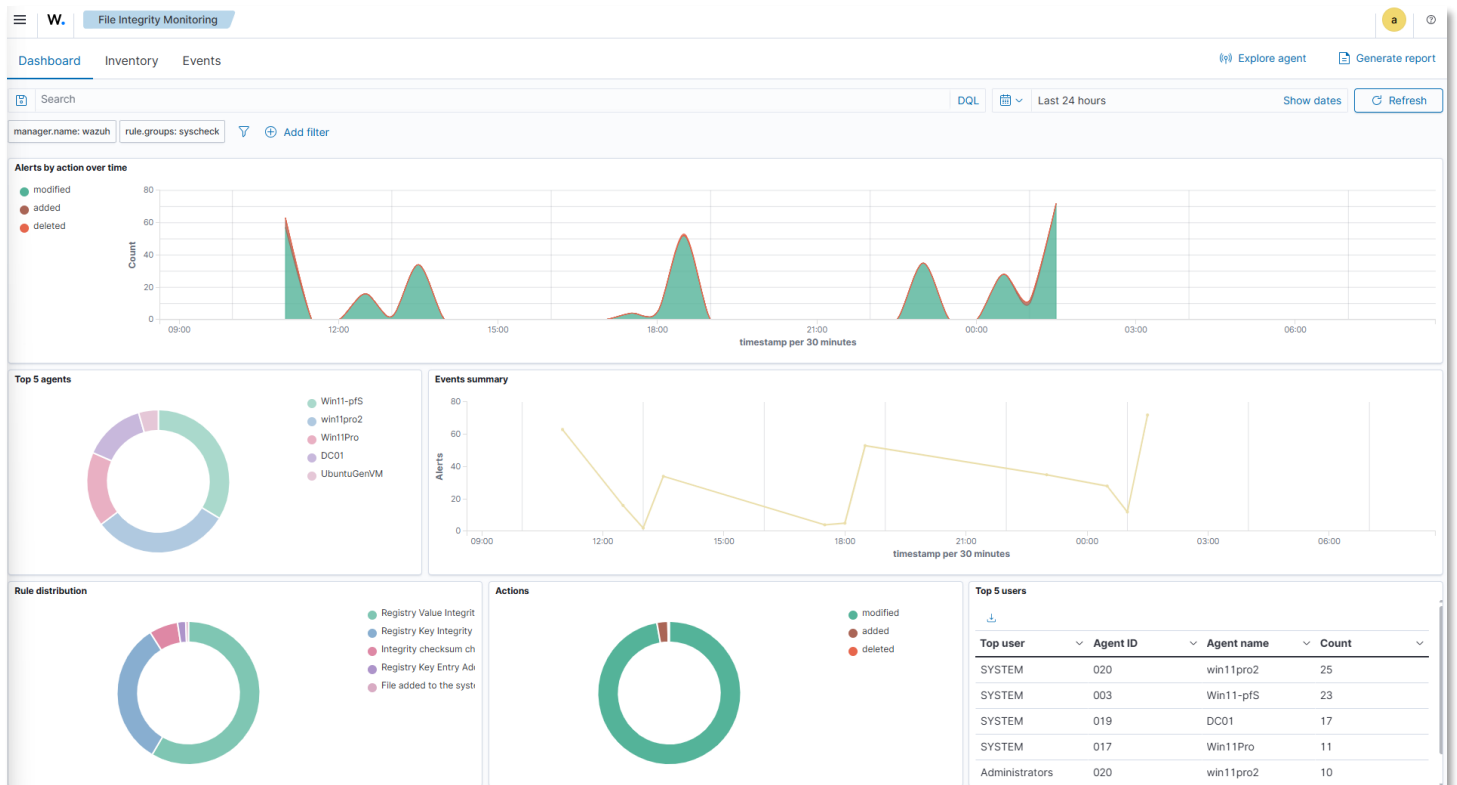
Triggered on new file creation in monitored directories or multiple failed login attempts

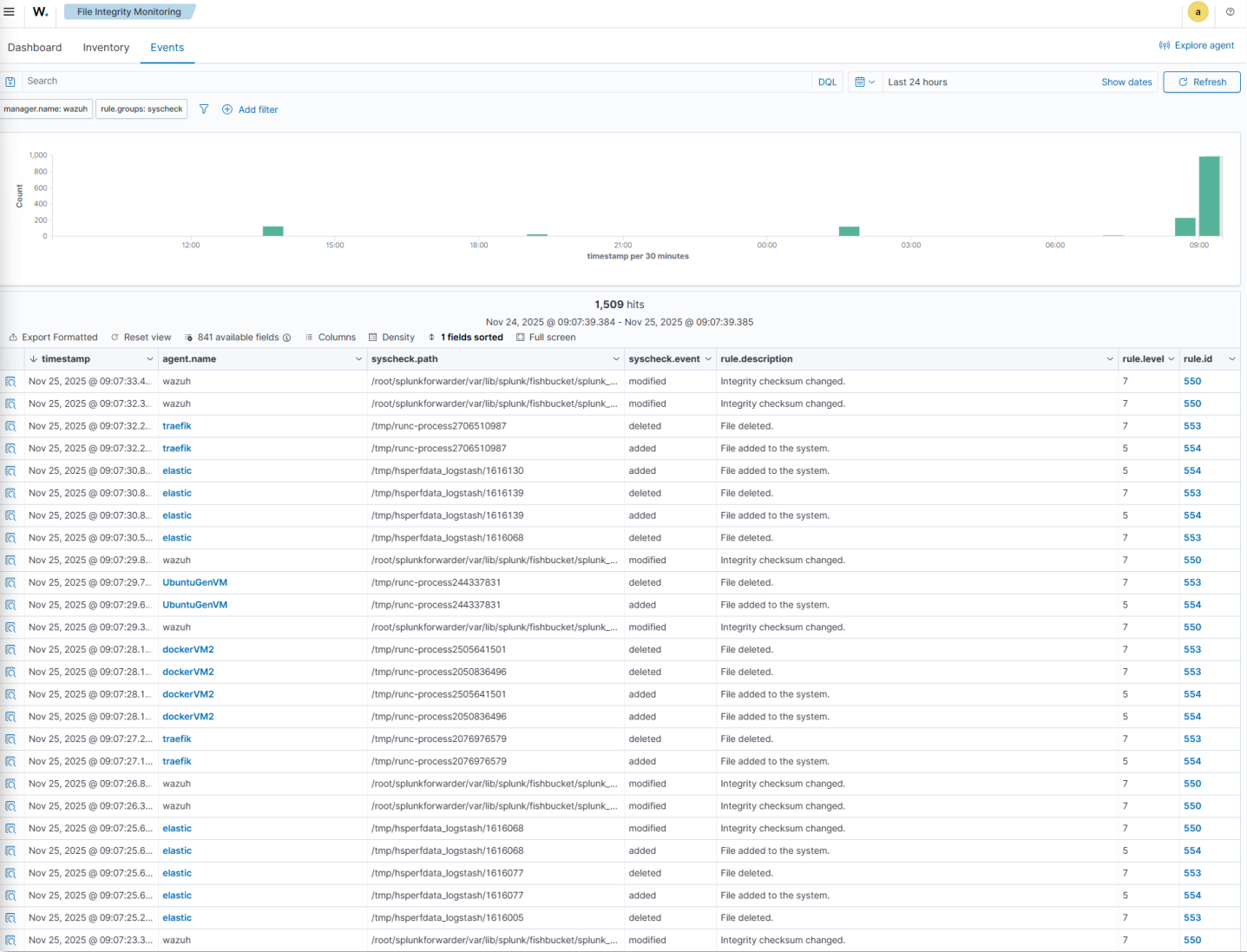
Wazuh Dashboard Analytics:

File Integrity Monitoring

Visualizations:

- Overview:
 - Alerts by action over time: Line chart showing file modifications, additions and deletions
 - Top 5 agents: Donut chart showing the top 5 agents by events
 - Rule Distribution: Donut chart showing the events by rule type
 - Actions: Donut chart showing events by modification, addition or deletion action
 - Top 5 Users: table showing the number of events by agent ID
- Events:
 - Events over time
 - Event Summary table showing total events by agent, file, action and rule information
- Inventory:
 - Detailed inventory information showing file and registry (Windows) actions including path/key, modification timestamp, user, file size.





≡	W.	File Integrity M...	DC01
Dashboard	Inventory	Events	
Files (11)	Windows Registry (6391)		
Registry (6,391)			
Search			
Registry ↑			Last modified ▲
HKEY_LOCAL_MACHINE\Security			Nov 24, 2025 @ 10:15:29.000
HKEY_LOCAL_MACHINE\Security\Cache			Jul 31, 2025 @ 15:53:27.000
HKEY_LOCAL_MACHINE\Security\Policy			Jul 31, 2025 @ 21:06:29.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts			Nov 18, 2025 @ 13:04:23.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-1-0			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-1-0\ActSysAc			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-1-0\Privlgs			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-1-0\SecDesc			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-1-0\Sid			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-11			Jul 31, 2025 @ 15:52:53.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-11\ActSysAc			Jul 31, 2025 @ 15:52:53.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-11\Privlgs			Jul 31, 2025 @ 15:52:53.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-11\SecDesc			Jul 31, 2025 @ 15:52:53.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-11\Sid			Jul 31, 2025 @ 15:52:53.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-19			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-19\Privlgs			Jul 30, 2025 @ 14:26:32.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-19\SecDesc			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-19\Sid			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-20			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-20\Privlgs			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-20\SecDesc			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-20\Sid			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-32-544			Jul 30, 2025 @ 14:26:27.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-32-544\ActSysAc			Jul 30, 2025 @ 14:26:32.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-32-544\Privlgs			Jul 31, 2025 @ 15:52:53.000
HKEY_LOCAL_MACHINE\Security\Policy\Accounts\S-1-5-32-544\SecDesc			Jul 30, 2025 @ 14:26:27.000

Configuration Management

Visualizations:

- Dashboard:
 - Per agent SCA showing a summary of passed, failed and N/A results and overall score
 - Policy information, scan data and checks
 - Inventory information: policy, description, scan data, score
 - Events: Event count over time, detailed check and policy information

W.

Configuration A...

fedora

Dashboard

Inventory

Events

fedora (033)

CIS DISTRIBUTION INDEPENDENT LINUX BENCHMARK ...

Passed (95)

Failed (90)

Not applicable (5)

CIS Distribution Independent Linux Benchmark v2.0.0.

Passed

95

Failed

90

Not applicable

5

Score

51%

End scan

Nov 24, 2025 @ 01:44:37.000

Checks (190)

Refresh

Export formatted

Search

WQL

ID ↑	Title	Target	Result	
36000	Ensure mounting of cramfs filesystems is disabled.	Command: modprobe -n -v cramfs,lsmod	Passed	▼
36001	Ensure mounting of freevxfs filesystems is disabled.	Command: modprobe -n -v freevxfs,lsmod	Passed	▼
36002	Ensure mounting of jffs2 filesystems is disabled.	Command: modprobe -n -v jffs2	Failed	▼
36003	Ensure mounting of hfs filesystems is disabled.	Command: modprobe -n -v hfs	Failed	▼
36004	Ensure mounting of hfsplus filesystems is disabled.	Command: modprobe -n -v hfsplus	Failed	▼
36005	Ensure mounting of squashfs filesystems is disabled.	Command: modprobe -n -v squashfs	Failed	▼
36006	Ensure mounting of udf filesystems is disabled.	Command: /sbin/modprobe -n -v udf	Failed	▼
36007	Ensure mounting of FAT filesystems is limited.	Command: /sbin/modprobe -n -v vfat	Failed	▼
36008	Ensure /tmp is configured.	Command: findmnt -kn /tmp	Passed	▼
36009	Ensure nodev option set on /tmp partition.	Command: findmnt -kn /tmp	Passed	▼
36010	Ensure nosuid option set on /tmp partition.	Command: findmnt -kn /tmp	Passed	▼
36011	Ensure noexec option set on /tmp partition.	Command: findmnt -kn /tmp	Failed	▼
36012	Ensure separate partition exists for /var.	Command: findmnt -kn /var	Failed	▼
36013	Ensure separate partition exists for /var/tmp.	Command: findmnt -kn /var/tmp	Failed	▼
36014	Ensure nodev option set on /var/tmp partition.	Command: findmnt -kn /var/tmp	Failed	▼
36015	Ensure nosuid option set on /var/tmp partition.	Command: findmnt -kn /var/tmp	Failed	▼
36016	Ensure noexec option set on /var/tmp partition.	Command: findmnt -kn /var/tmp	Failed	▼
36017	Ensure separate partition exists for /var/log.	Command: findmnt -kn /var/log	Failed	▼
36018	Ensure separate partition exists for /var/log/audit.	Command: findmnt -kn /var/log/audit	Failed	▼
36019	Ensure separate partition exists for /home.	Command: findmnt -kn /home	Failed	▼
36020	Ensure nodev option set on /home partition.	Command: findmnt -kn /home	Failed	▼
36021	Ensure nodev option set on /dev/shm partition.	Command: findmnt -kn /dev/shm	Passed	▼
36022	Ensure nosuid option set on /dev/shm partition.	Command: findmnt -kn /dev/shm	Passed	▼

W.

Configuration A...

fedora

Dashboard

Inventory

Events

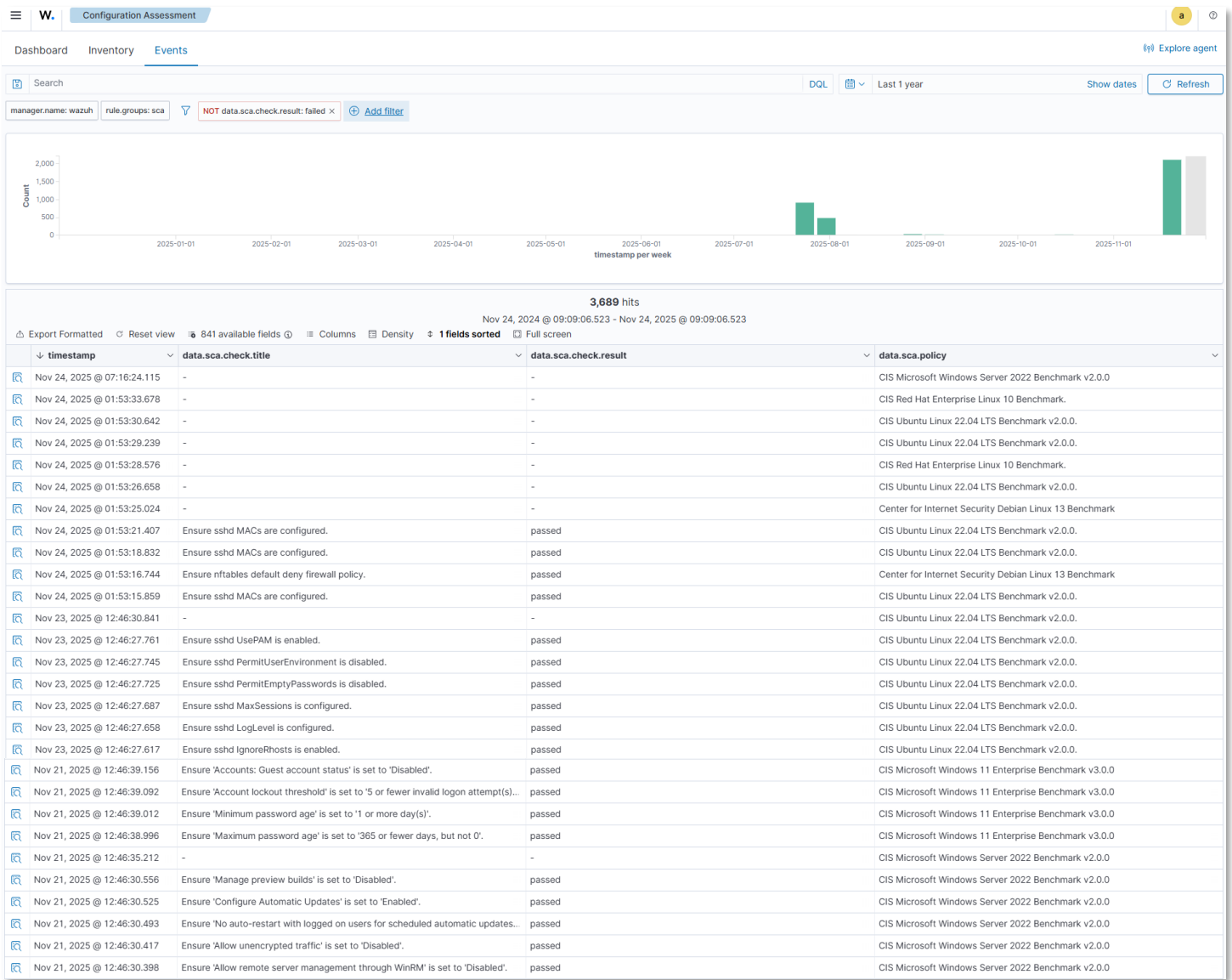
fedora (033)

Policy	Description	End scan	Passed	Failed	Not applicable	Score
CIS Distribution Independent Linux Benchmark v2.0.0.	This document provides prescriptive guidance for establishing a secu...	Nov 24, 2025 @ 01:44:37.000	95	90	5	51%

Rows per page: 15 < 1 >

Policy	Description	End scan	Passed	Failed	Not applicable	Score
CIS Red Hat Enterprise Linux 10 Benchmark.	This document provides prescriptive guidance for establishing a secu...	Nov 24, 2025 @ 01:44:44.000	78	74	7	51%

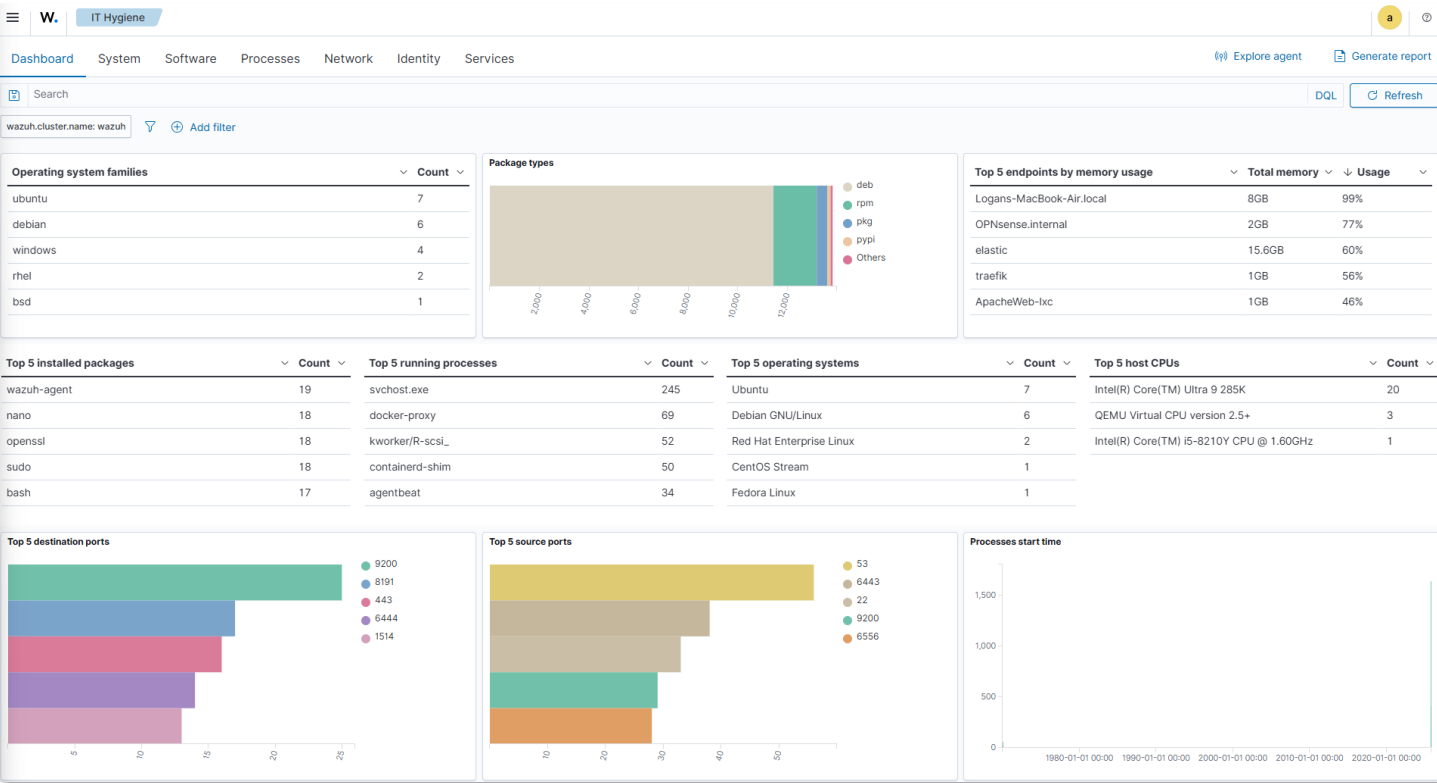
Rows per page: 15 < 1 >

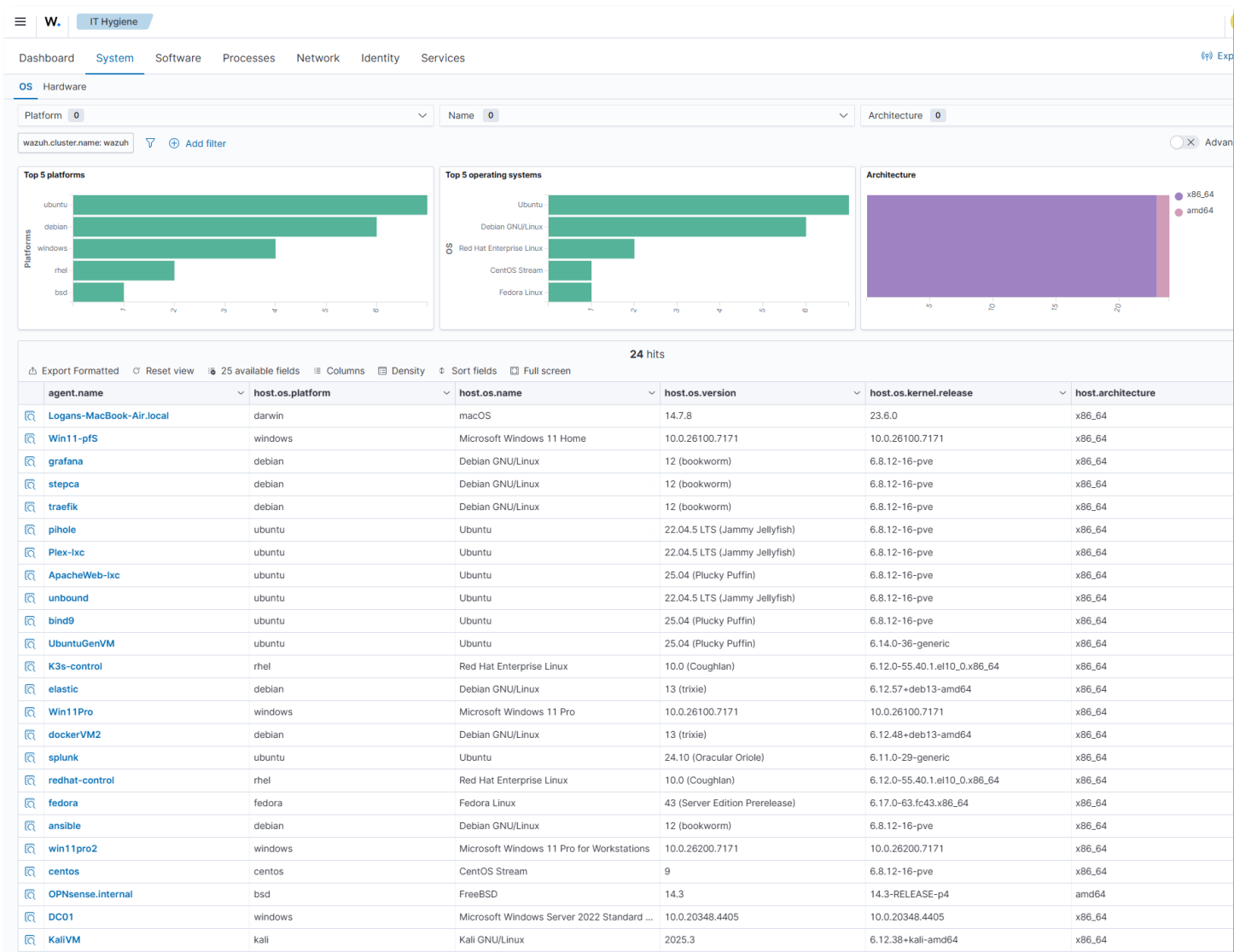


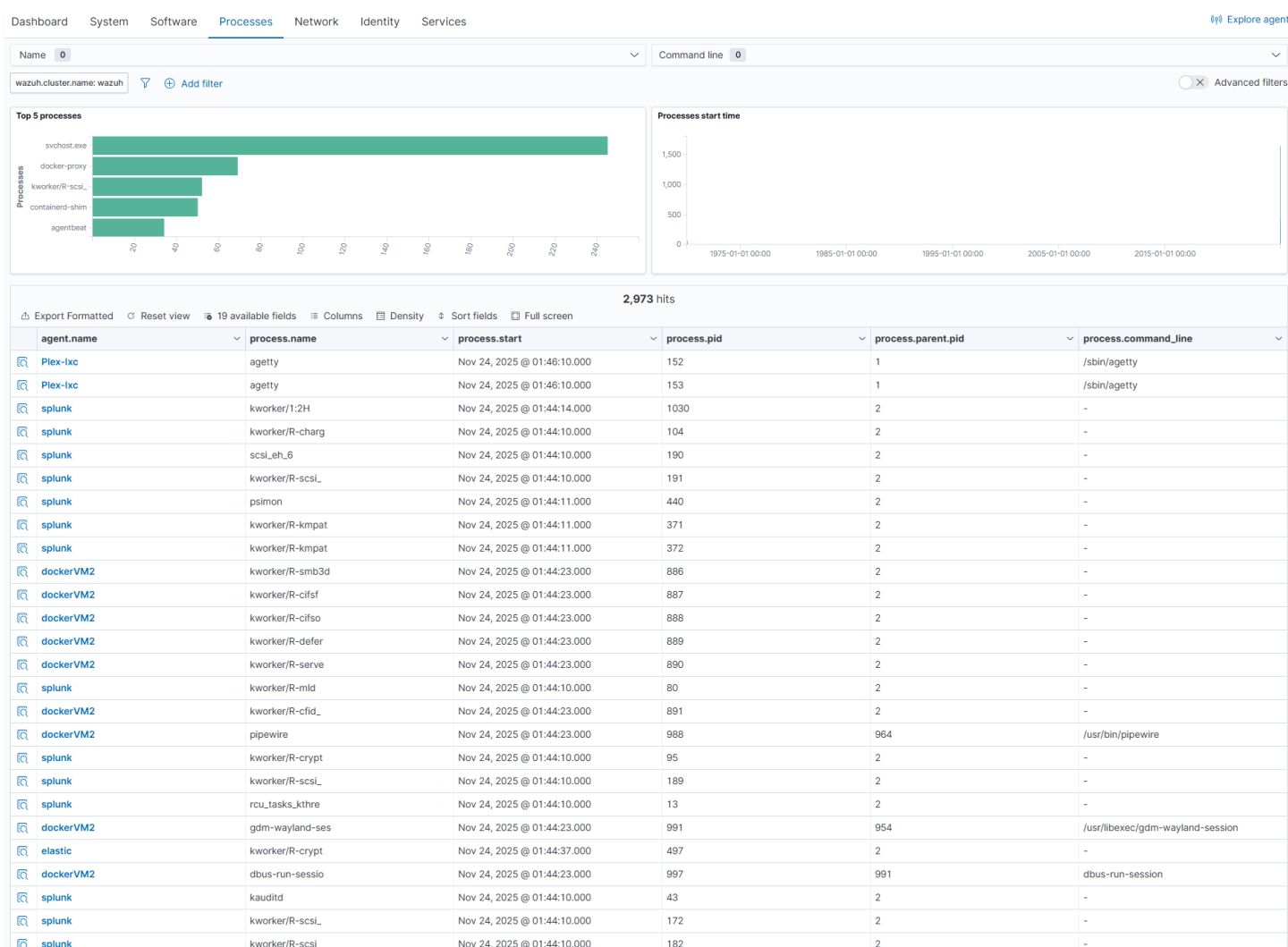
Security Operations / IT Hygiene Dashboard:

Visualizations:

- Dashboard: Overview of deployed agent hygiene showing OS, installed packages, running processes, hardware, network traffic and identity information.
 - System: Platform, OS and architecture information
 - Software: Package, KB, Browser extension and vendor information
 - Processes: Process name, start,PID, PPID and command line information
 - Network: IP address, interface, protocol, listener and traffic information.
 - Identity: User and group, skell and home directory information
 - Services: Running service information







Threat Intelligence:

- Threat Hunting
 - Total threats detected, alert level, authentication failures/successes, attacks by MITRE tactic/technique, alerts by agent.
- Vulnerability Detection:
 - Severity level counts, top 5 CVEs, vulnerabilities by OS, agent, package and year.
- MITRE ATT&CK
 - Alert evolution over time, top tactics, attacks by technique, tactics and techniques by agent.

Threat Hunting Queries:

Suspicious PowerShell Execution:

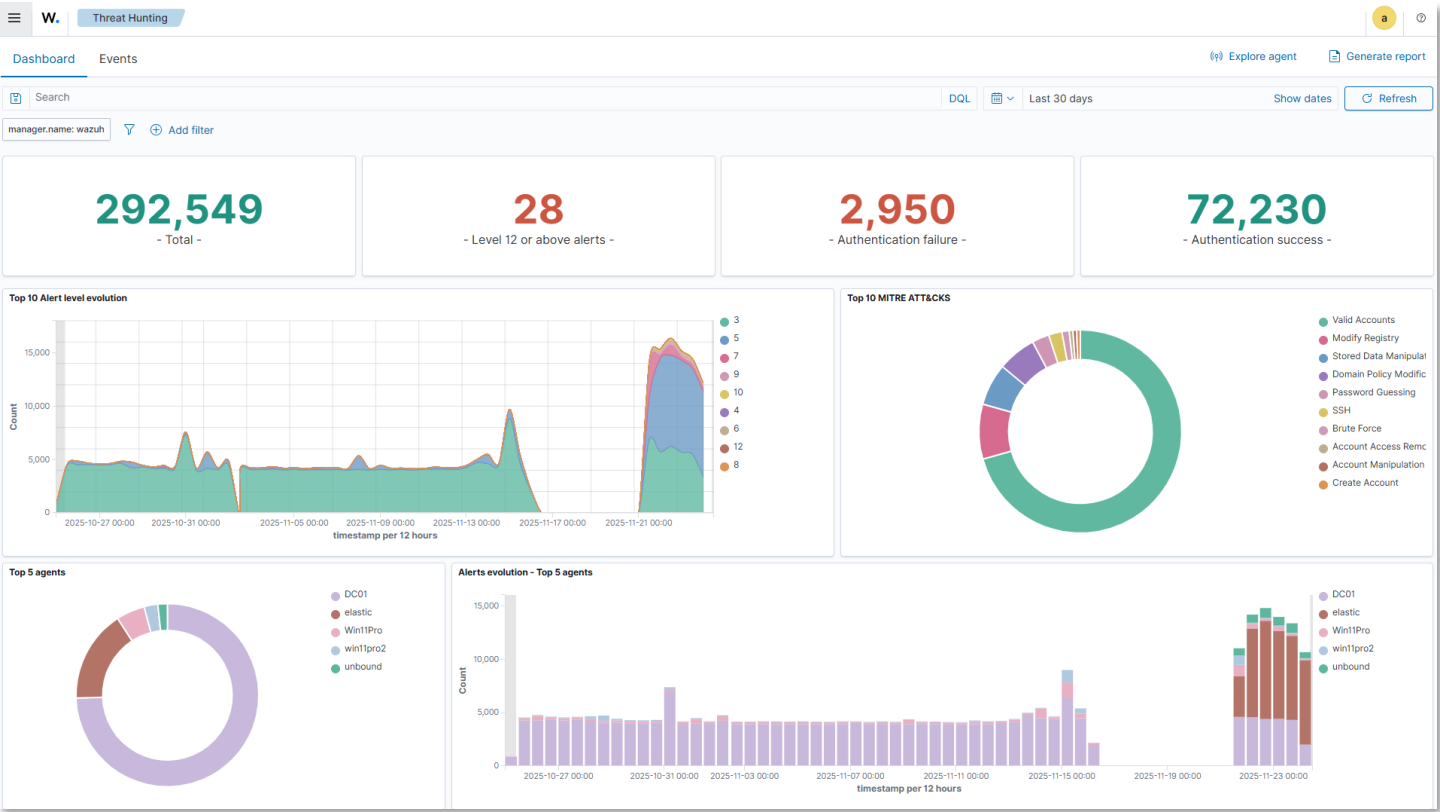
```
rule.groups: windows AND data.win.eventdata.image: *powershell.exe
AND (data.win.eventdata.commandLine: *-encodedcommand*
OR data.win.eventdata.commandLine: *bypass*)
```

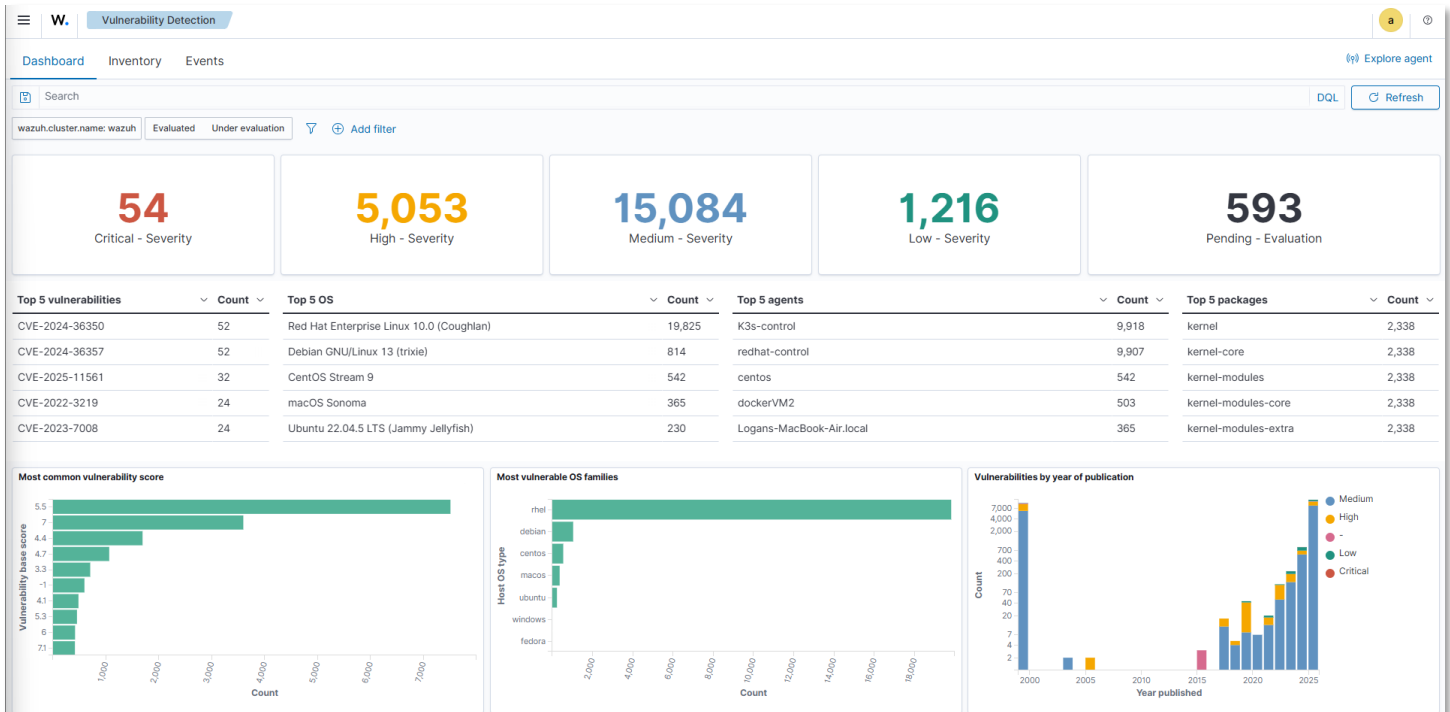
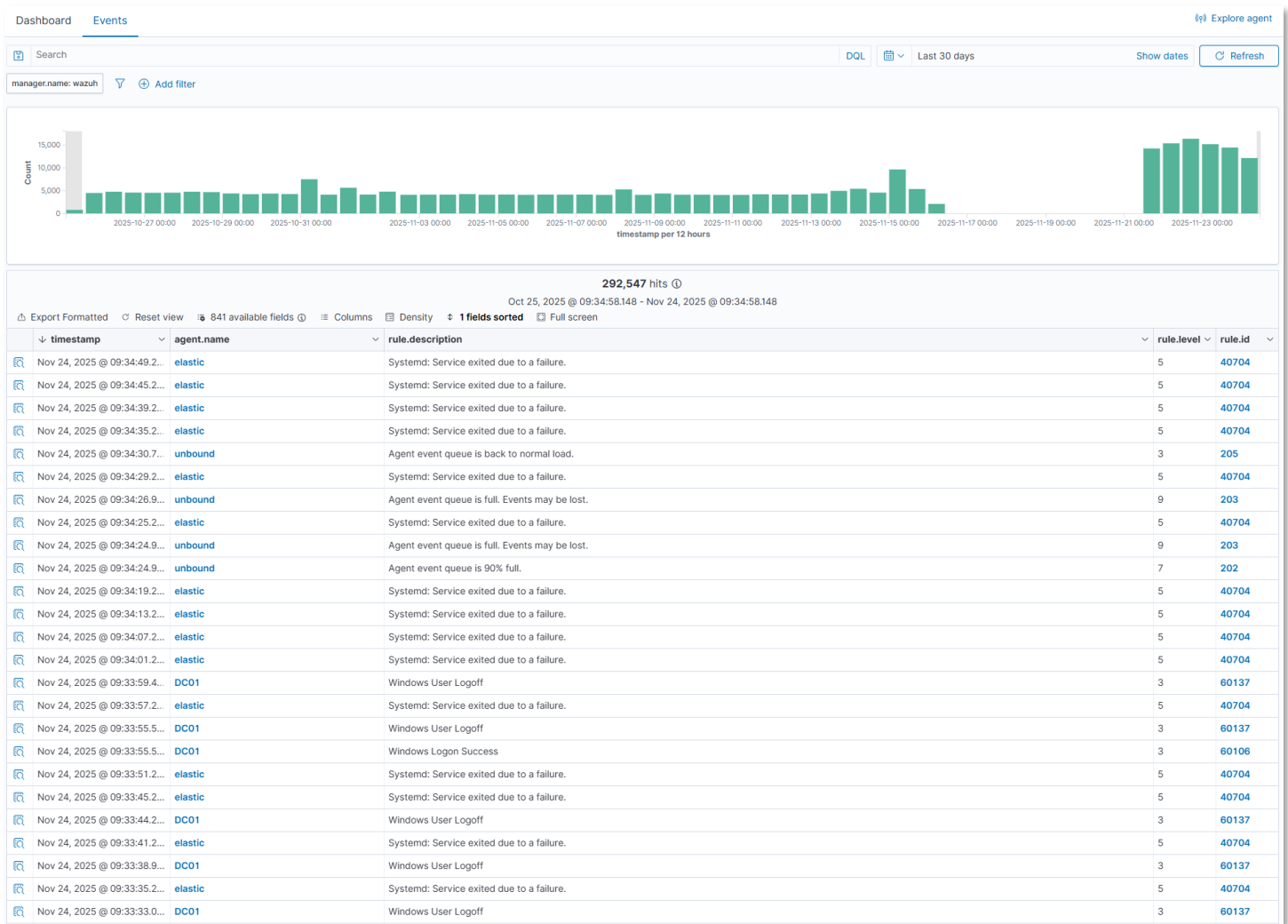
Lateral Movement Detection:

```
rule.id: 60204 AND rule.description: Windows: User logon with explicit credentials
| stats count by data.win.eventdata.subjectUserName, agent.name
```

Privilege Escalation Attempts:

```
rule.groups: authentication_success AND data.win.eventdata.targetUserName: Administrator
| where data.win.eventdata.subjectUserName != Administrator
```





DashboardInventoryEvents

Search

QDLRefresh

wazuh.cluster.name: wazuhEvaluatedUnder evaluationAdd filter

22,000 hits

Export FormattedReset view48 available fieldsColumnsDensitySort fieldsFull screen

agent.name	package.name	package.version	vulnerability.description	vulnerability.severity	vulnerability.id
redhat-control	systemd-udev	257-9.el10_0.1	A vulnerability was found in systemd-cored...	Medium	CVE-2025-4598
redhat-control	microcode_ctl	4:20250211-1.el10_0	Insufficient resource pool in the core mana...	Medium	CVE-2025-20103
redhat-control	microcode_ctl	4:20250211-1.el10_0	Uncaught exception in the core managem...	Medium	CVE-2025-20054
redhat-control	microcode_ctl	4:20250211-1.el10_0	Improper input validation in UEFI firmware f...	Medium	CVE-2024-28047
redhat-control	microcode_ctl	4:20250211-1.el10_0	Improper Finite State Machines (FSMs) in ...	Medium	CVE-2024-31068
redhat-control	microcode_ctl	4:20250211-1.el10_0	Improper restriction of software interfaces ...	Medium	CVE-2024-48869
redhat-control	microcode_ctl	4:20250211-1.el10_0	Improper finite state machines (FSMs) in th...	Medium	CVE-2024-21853
redhat-control	microcode_ctl	4:20250211-1.el10_0	Improper access control in the EDECCSSA ...	Medium	CVE-2024-36293
redhat-control	python3-pip-wheel	23.3.2-7.el10	A flaw was found in urllib3. The library fails ...	Medium	CVE-2025-50182
redhat-control	python3-pip-wheel	23.3.2-7.el10	A flaw was found in urllib3. The 'PoolMana...	Medium	CVE-2025-50181
redhat-control	sssd-krb5-common	2.10.2-3.el10_0.2	A flaw was found in the integration of Activ...	High	CVE-2025-11561
redhat-control	libxslt	1.1.39-8.el10_0	A flaw was found in the libxslt library. The s...	High	CVE-2025-7424
redhat-control	libxslt	1.1.39-8.el10_0	A flaw was found in the exsltFuncResultCo...	Low	CVE-2025-11731
redhat-control	libxslt	1.1.39-8.el10_0	A flaw was found in libxslt package. Proces...	Medium	CVE-2023-40403
redhat-control	libxslt	1.1.39-8.el10_0	A flaw was found in libxslt where the attrib...	High	CVE-2025-7425
redhat-control	libxslt	1.1.39-8.el10_0	A use-after-free vulnerability was found in L...	Medium	CVE-2025-10911
redhat-control	python3-rpds-py	0.17.1-6.el10	A flaw was found in PyO3. This vulnerabil...	Medium	CVE-2024-9979
redhat-control	amd-gpu-firmware	20250812-15.7.el10_0	A transient execution vulnerability in some ...	Medium	CVE-2024-36357
redhat-control	amd-gpu-firmware	20250812-15.7.el10_0	A transient execution vulnerability in some ...	Medium	CVE-2024-36350
redhat-control	sssd	2.10.2-3.el10_0.2	A flaw was found in the integration of Activ...	High	CVE-2025-11561
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	Medium	CVE-2025-40206
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	Medium	CVE-2025-40201
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	Medium	CVE-2025-40200
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	High	CVE-2025-40199
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	Medium	CVE-2025-40198
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	Medium	CVE-2025-40196
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	Medium	CVE-2025-40195
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	Medium	CVE-2025-40194
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	Medium	CVE-2025-40187
redhat-control	kernel-tools-libs	6.12.0-55.40.1.el10_0	In the Linux kernel, the following vulnerabi...	Medium	CVE-2025-40181

